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Prova finale

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”Positron Emission Tomography”

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1 Introduction

The aim of this project is to illustrate the operating principles behind a PET (Positron Emission Tomography) detection system, to study image reconstruction, to evaluate image quality, and to simulate the system with some simplifications.

2 PET Description

PET is a medical imaging technique capable of measuring the spatial concentration of radionuclides present in the patient after the administration of a radiopharmaceutical, with a sensitivity of $10^{-11}/10^{-12}$ mol/L. This technology exploits the emission of two γ rays following the annihilation of a positron, previously generated by the β^+ decay of a radioisotope.

2.1 Radiopharmaceutical

In the diagnosis and management of oncological patients, positron emission tomography (PET) with 18F-fluorodeoxyglucose (FDG) is considered a diagnostic tool of great importance.

The method is based on the administration, generally intravenously, of a radiopharmaceutical, namely biologically active molecules labelled with radionuclides, followed by the evaluation of its distribution within organs and tissues.

In PET imaging, radionuclides that decay through positron emission (β^+) from the nucleus must be used. The radiopharmaceuticals used are therefore characterized by the following properties:

- High radioactivity, resulting from β^+ decay;
- Short half-life of the radioisotopes, which also implies a reduction in the dose delivered to the patient undergoing the examination.

One of the most commonly used radionuclides in PET analysis is Fluorine-18 (18F). 18F is a positron-emitting radioisotope with a half-life of 110 minutes. It is used to label glucose, forming the radiopharmaceutical 18F-FDG, which is able to localize in regions where cellular metabolism is highly active. The greatest accumulation of the radiopharmaceutical occurs in tissues with intense glycolytic activity: tumour cells, the grey matter of the brain (9% of the administered activity accumulates in this organ), and the myocardium under specific metabolic conditions (3%).

In the following sections, particular attention will be given to the analysis of a brain PET case.

2.2 Physics

The positron, which will be denoted in this report as β^+ , is the antiparticle of the electron and is generated by the decay of particles with an excess number of protons:

$$X_Z \rightarrow X_{Z-1}^* + \beta^+ + \nu_e$$

In some cases, the resulting nucleus remains excited, indicated by the superscript “*”, and subsequently reaches stability by emitting a single γ ray.

An important piece of information is the maximum energy of the positron: this can range from hundreds of keV to a few MeV, depending on the radioisotope. The energy defines the positron “range”, namely the distance travelled before annihilation. Once emitted, the positron collides with electrons, considerably changing its direction. Following these collisions, when its energy becomes sufficiently low, it annihilates with an electron, emitting two γ rays “back to back”, therefore at an angle of 180° (with an error of $\pm 0.5^\circ$, using water as an approximation of human tissue), each with an energy of 511 keV.

2.3 Instrumentation

The PET system is based on a set of detectors surrounding the patient. The minimum number of detectors is two, which is necessary in order to detect two back-to-back rays. If only two detectors were used, they would also need to rotate around the patient.

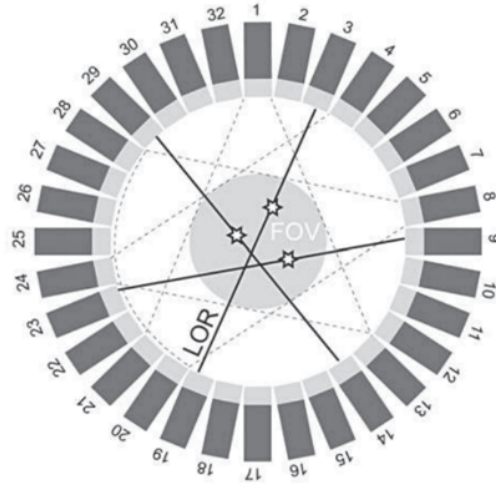


Figura 1: If two rays are detected at an angle of $180^\circ \pm 0.5^\circ$ and separated by a time interval Δt (the “time window”, typically equal to 10 ns), the rays are correlated and the LOR (line of response) is stored. The LOR represents the line along which the annihilation occurred. It is also possible to define the field of view of a detector, namely the region of space that contains all and only the LORs detectable by that detector.

2.3.1 Multi-ring

Today, these “rotating detectors” are no longer used. Instead, a ring made up of multiple detectors is preferred, so that the system does not need to rotate. Each detector is placed in coincidence with a diametrically opposite arc.

The system is able to analyse a section whose thickness is equal to the detector thickness. Therefore, by increasing the number of rings, the field of view can be expanded.

It is also possible to use multiple rings (“multi-ring”), choosing between 2D and 3D methods.

In the 2D method, several images are obtained on parallel planes, without setting any coincidence between different rings. This method is less efficient, but incorrect LORs are not recorded.

The 3D method, instead, exploits coincidences between different rings, obtaining LORs located on planes that are not necessarily parallel. This method would allow a better image to be reconstructed, but it may store incorrect LORs due to possible scattering events, as shown in the figure.

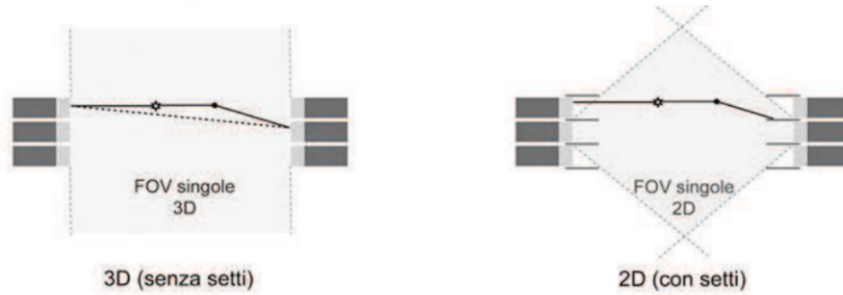


Figura 2: 3D and 2D methods.

2.3.2 Detectors

A detection event is recorded if two opposite detectors (or detectors within the error range of $\pm 0.5^\circ$) both register a signal within the time interval defined by the coincidence time window. The detector pair must be able to provide both the position and the energy of the interaction.

The most commonly used technologies are scintillators coupled to a position-sensitive photo-detection system (see Figure 3).

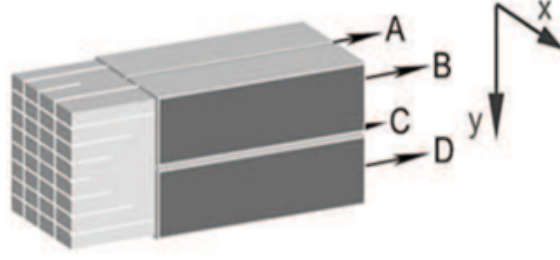


Figura 3: Block detector.

The most common solution is the “block detector”: a block of bismuth germanate ($Bi_4Ge_3O_{12}$). Longitudinal cuts are made in the block, gradually becoming deeper closer to the edge. These cuts are filled with a reflective material in order to limit the diffusion of the light signal and Compton scattering.

The block is connected to a 2×2 matrix of photomultipliers. The light reaches one of the four photomultipliers proportionally to the interaction position: the light signal spreads, so if, for example, it originates on the edge between A and B, these photomultipliers will receive a stronger signal than C and D.

The maximum precision of this instrument is proportional to the size of the photomultipliers, and the position is obtained as the energy-weighted average of the signal detected by the four elements.

$$X = \frac{(SA + SC) - (SB + SD)}{E}$$

$$Y = \frac{(SA + SB) - (SC + SD)}{E}$$

(X, Y) : coordinates;

SA, SB, SC, SD : signals produced by the respective photomultipliers;

E : interaction energy, $E = SA + SB + SC + SD$.

2.3.3 Detection Crystals

Today, bismuth germanate is no longer commonly used. Instead, lutetium oxyorthosilicate (LSO), lutetium-yttrium oxyorthosilicate (LYSO), and gadolinium oxyorthosilicate (GSO) are preferred. In addition, the scintillator itself is subdivided into elements with a minimum size of 4.4 mm.

Detection crystals must satisfy the following requirements:

- High light output, which improves signal encoding, allows smaller elements to be produced and therefore increases precision.
- High electron density, which increases efficiency for the same thickness while reducing parallax error. This error depends on how deep inside the crystal the γ ray is detected.
- Short decay constant τ , so that a smaller time window can be used, reducing the number of random events detected.

2.4 Time of Flight (TOF)

With traditional PET, for each event only the LOR is known. However, thanks to technological developments, it is possible to improve the precision of the measurement. In fact, by knowing the velocity of the γ rays ($V_\gamma = 3 \cdot 10^8 m/s$) and measuring the time difference between the two detected signals, it is possible to determine, within a certain uncertainty, the point on the LOR where the annihilation occurred.

The uncertainty ΔS , measured as FWHM (Full Width at Half Maximum), is given by half the distance travelled by the rays during the time uncertainty: $\Delta S = \frac{c \times \Delta t}{2}$.

The technique that exploits these properties is called TOF-PET and will be described in greater detail later.

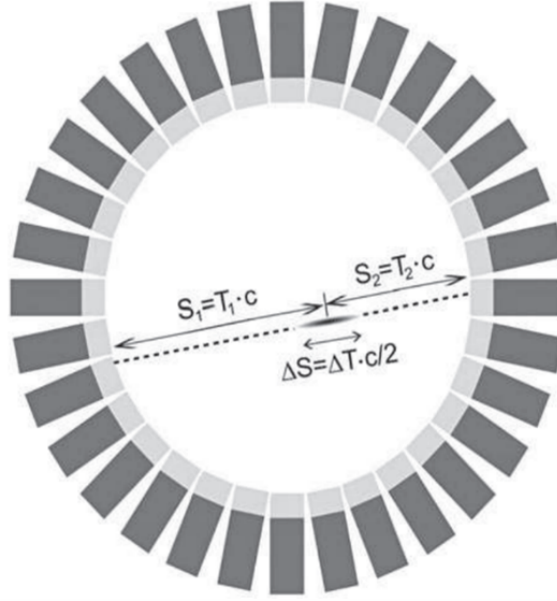


Figure 4: Time of Flight

2.5 Image Reconstruction

The LORs of the events considered useful are stored in 3D using the zenith and azimuthal angles as parameters.

Analytical or iterative algorithms are used to reconstruct the images.

Analytical algorithms: a filtered back-projection is applied to the obtained results. This consists of superimposing multiple views and filtering them in order to obtain a sharper image. This method is simple and fast, but it leaves a star-shaped artefact, shown below.



Figure 5: The star-shaped artefact is caused by the intersection of multiple lines all passing through the same point.

Iterative algorithms: these are based on statistical methods. They use estimates of the distribution in order to reduce noise and avoid the formation of star-shaped artefacts, but they are more complex and slower.

2.6 Spatial Resolution

Spatial resolution is limited by several errors, due both to the limitations of the available technology and to the physics of the phenomenon. The best parameter for evaluating spatial resolution is the FWHM.

$$FWHM = a\sqrt{\left(\frac{d}{2}\right)^2 + b^2 + (0.0022D)^2 + r^2 + p^2}$$

The parameters appearing in the formula are associated with the effects described below:

Coefficient (a): experimental factor that depends on the reconstruction method.

Range effect (r): the objective of the measurements would be to localize the positron emission point. However, it is only possible to measure where the positrons annihilate. The distance travelled between these two instants introduces an error known as the range effect. (1 in Figure 6).

Angular deviation (0.0022D): the two γ rays are not emitted exactly at 180° , but at $180^\circ \pm 0.5^\circ$. (2 in Figure 6).

Encoding error (b) & (d/2): this is due to the block detector. If a γ ray undergoes Compton interactions before producing a photoelectric event, the crystal will detect the position as the centre of mass of the energy released during the interactions. (1 in Figure 7).

Parallax error (p): this is due to the lack of information associated with the depth of interaction inside the scintillator. If the interaction occurs deep inside the crystal or near its surface, it is not possible to distinguish between the two cases. This effect becomes more pronounced farther from the axis. (2 in Figure 7).

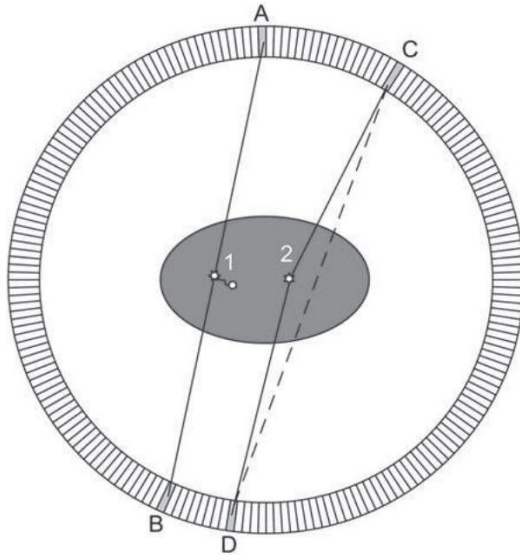


Figura 6: 1 range effect. 2 angular deviation.

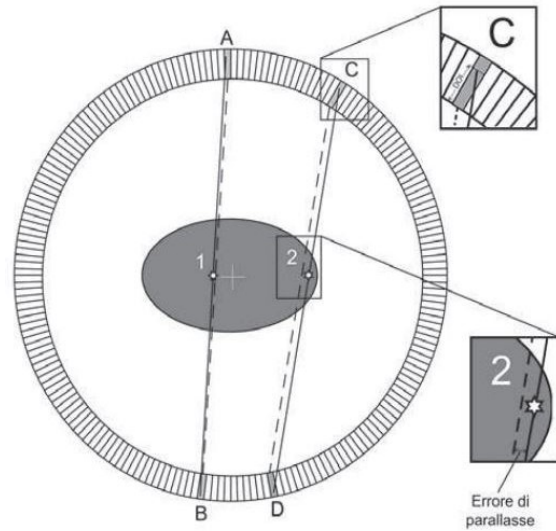


Figura 7: 1 encoding error. 2 parallax.

2.7 Noise

A recorded event can be true, scattered, or random:

True event: corresponds to two γ rays that have not been deflected by Compton scattering.

Scattered event: one of the two γ rays has undergone a deflection due to Compton scattering.

Random event: two γ rays are detected within the coincidence time window, but they do not originate from the same annihilation event.

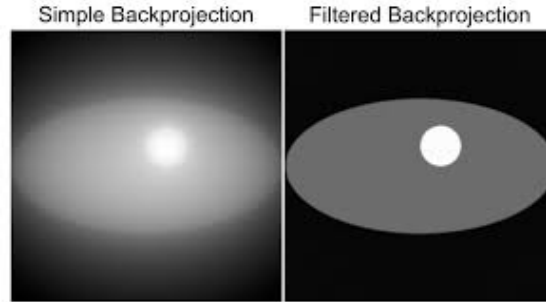
To identify scattered or random events, an estimate of the expected distribution and a selection based on the detected energy can be used.

3 Image Reconstruction Methods

3.1 Filtered Back-Projection (FBP)

Filtered Back-Projection (FBP) is an analytical method for tomographic reconstruction. It is the most commonly used method for solving inverse tomographic problems, either as a complete reconstruction method or as an initial “guess” (zero-order solution) for iterative methods.

It consists of a back-projection of the data, preceded by a high-pass frequency filter. In fact, after a simple Back Projection, the resulting image is blurred because it is obtained through the direct intersection of the LORs. Therefore, FBP is applied to make the image sharper. In practice, it increases the image contrast.



3.2 Maximum-Likelihood Expectation Maximization (MLEM)

Modern PET systems use iterative image reconstruction algorithms such as Maximum-Likelihood Expectation-Maximization (MLEM).

Reconstructing a PET image means finding the number of counts in each pixel of the image, knowing the counts measured for all pairs of detectors in coincidence.

Following an annihilation event, we know that two photons are generated “back to back”.

As mentioned above, if both photons are detected by two detectors in opposite positions, within a small uncertainty and within a short time window, they are said to be in coincidence. Each pair generates a Line of Response (LOR), along which the annihilation event occurred.

Through the intersection of multiple LORs, it is possible to estimate the approximate position where the annihilation events occurred.

For each generated line, we store as parameters the orthogonal distance from the centre and the angle of inclination.

Using these two parameters, together with a sufficiently large number of LORs, we obtain the sinogram, a graph that relates the angles to the distances from the centre.

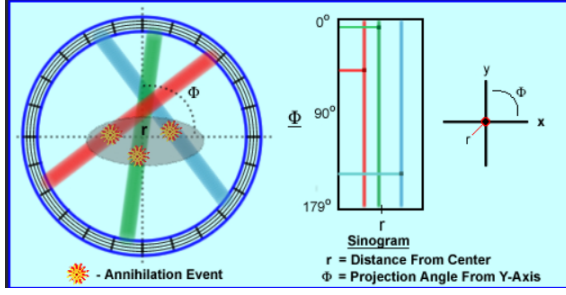


Figura 8: Characteristic parameters for the sinogram.

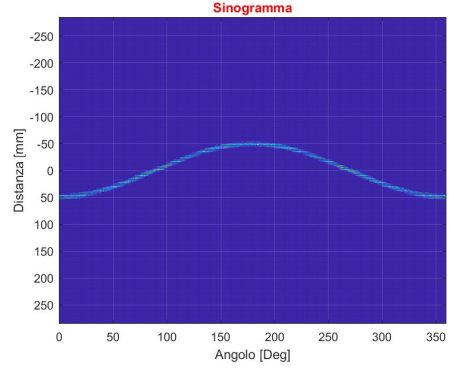


Figura 9: Sinogram obtained with a source located 5 cm from the centre, using a ring with 360 detectors.

To move from the measured data to the sinogram, the Radon transform is used¹.

Through the inverse Radon transform², instead, the sensitivity image, or sensitivity map, is calculated. This expresses how many back-projection lines pass through each pixel and therefore the contribution of each pixel to the total sum of counts in the detectors, taking into account the system geometry and the detector characteristics.

The actual iterative method is then applied.

MLEM is based on the calculation of a new value for each pixel of the image, starting from the image values obtained at the previous iteration.

```
% Codice MATLAB per il metodo MLEM
function final_image=reconstruct_image(sinogram,angles,number_iteration,dim_im)

sing=ones(size(sinogram));

sens_im=iradon(sing,angles,'none',dim_im);
final_image=ones(dim_im);
for i=1:number_iteration

    foreward_projection=radon(final_image,angles);
    ratio=sinogram./(foreward_projection+0.000000001);
    correction=iradon(ratio,angles,'none',dim_im)./sens_im;
    final_image=final_image.*correction;
end
end
```

A sinogram is initialized, in this case as a matrix of ones with the same size as the measured sinogram. The sensitivity image of this initialized sinogram is then defined. These two terms are only used to calculate the correction applied to the final image.

An initial image is then initialized, also as a matrix of ones. At each iteration, the Radon transform is computed, thus obtaining a sinogram that changes at every iteration. A pixel-by-pixel ratio is then calculated between the measured sinogram and the sinogram obtained at each iteration. Subsequently, the back-projection lines of this ratio are computed in order to obtain the correction to be applied to the image.

¹The Radon transform is an integral transform that maps a function f , defined on a plane, to a function defined on the two-dimensional space of the lines of that plane. It is defined as:

$R_\theta(\rho)[f(x,y)] = \iint_D f(x,y) \cdot (\delta)(\rho - x\cos\theta - y\sin\theta) dx dy$, where $D = (-\infty, +\infty) \times (-\infty, +\infty)$.

²The inverse Radon transform is defined as: $(\frac{1}{2\pi}) \int_{-\infty}^{+\infty} (\frac{d}{dy}) H[U(m, y - mx)] dm$.

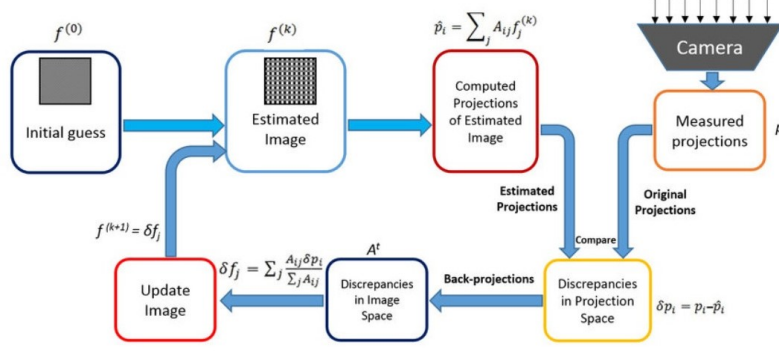


Figura 10: Block diagram of the MLEM method.

3.3 Time of Flight (TOF)

The Time of Flight (TOF) technique is another imaging technique.

It measures the time taken by the photons to travel the distance between the annihilation point and the detectors. In conventional imaging, the annihilation of a positron is recorded as a line of response (LOR) and does not provide the exact position of the event. Therefore, by calculating the intersection of multiple LORs, it is possible to infer the position where the annihilation occurred.

The TOF method, instead, allows us to measure the time interval between the detection of the photons in two detectors in coincidence. This makes it possible to identify more precisely the distance between the annihilation point and the centre of the detector ring, and therefore the position of the source.

The detector design is therefore critical for the TOF-PET system, and the average time resolution of scintillators currently available on the market is around 400÷600 ps. Other characteristics of the most common commercial solutions are reported in Tables 1, 2 and 3. A suitable scintillator must nevertheless have the following properties:

- high density and atomic number,
- short decay times,
- high light efficiency,
- transparency to the emitted light.

	Sodium Iodide NaI(Tl)	Barium Fluoride BaF ₂	Bismuth Germanate BGO	Lutetium Oxyorthosilicate LSO
Effective atomic number (Z)	51	54	74	66
Linear attenuation coefficient (cm ⁻¹)	0.34	0.44	0.92	0.87
Density (g/cm ³)	3.67	4.89	7.13	7.4
Light output (percentage relative to NaI(Tl))	100	5	15	75
Decay constant (ns)	230	0.8	300	40

Tabella 1: Characteristics of the main commercial scintillators.

	Gadolinium Oxyorthosilicate GSO	Lutetium-Yttrium Oxyorthosilicate LYSO	Lanthanum Bromide LaBr ₃
Effective atomic number (Z)	59	60	47
Linear attenuation coefficient (cm ⁻¹)	0.62	0.86	0.47
Density (g/cm ³)	6.7	7.1	5.3
Light output (percentage relative to NaI(Tl))	30	80	160
Decay constant (ns)	65	41	25

Tabella 2: Characteristics of the main commercial scintillators.

	Lutetium Fine Silicate LFS	Lutetium Orthoaluminate LuAP	Lutetium Iodide LuI ₃
Effective atomic number (Z)	63	65	60
Linear attenuation coefficient (cm ⁻¹)	0.82	0.9	~ 0.56
Density (g/cm ³)	7.3	8.34	5.6
Light output (percentage relative to NaI(Tl))	77	16	190
Decay constant (ns)	35	18	30

Tabella 3: Characteristics of the main commercial scintillators.

The code used for the implementation is reported below.

```
% MATLAB code for the TOF method
c = physconst('LightSpeed');
inctemp=1e-10;
prec=round(c*inctemp*100*scale/2);
x=-prec:prec;
mu=0;
sigma = prec / (2*sqrt(2*log(2)));
y = exp(-(x - mu).^2 / (2 * sigma^2));

image=zeros(D);
s=zeros(D);

for i=1:size(m,2)
    dis=round(c*(round(time(i,2),10)-round(time(i,1),10))*100/2)*scale;
    if abs(dis-prec)<R && abs(dis+prec)<R
        s(R+dis-prec:R+dis+prec,R+angASS_DIST(i,2))=s(R+dis-prec:R+dis+prec,R+
        ↪ angASS_DIST(i,2))+y';
        s=rotate_matrix_fast(s,angASS_DIST(i,1));
        image=image+s;
    end
    s=zeros(D);
end
```

From an operational point of view, in the implemented TOF method, the distance between the source and the centre of the LOR corresponding to the two detectors in coincidence is calculated using the formula³:

$$\frac{\Delta t_{detectors} \cdot c}{2}. \quad (1)$$

A Gaussian probability is then assigned to the points along the LOR. The point located at the centre position shifted by $\frac{\Delta t_{det} \cdot c}{2}$ is assigned an intensity equal to 1, while the other points are assigned an intensity such that they form a Gaussian distribution with FWHM (called **prec** in the code) equal to⁴:

$$\frac{\Delta t_{sensitivity} \cdot c}{2}, \quad (2)$$

and only if the dimensions are compatible with the geometry of the detector ring, the method updates the image by positioning the source.

³Where $\Delta t_{detectors}$ is the time discrepancy between the detections in the two detectors and c is the speed of light in vacuum ($c = 3 \times 10^8$ m/s).

⁴The introduction of an error was made to compensate for the infinite time resolution of the detectors used in the simulation, in order to make the model more realistic.

4 The Simulation

The PET system was reconstructed in a virtual environment using two programs: FLUKA for the simulation of the physical behaviour of particles and data collection, and MATLAB for data processing, image reconstruction and analysis. Due to computational limitations, some simplifications were introduced, which will be listed below.

4.1 Environment

Data collection was performed using a Monte Carlo code (FLUKA), which recreates a simulated environment. In this environment it is possible to generate particles, called primaries, by defining their initial position, direction and energy. These particles are created sequentially, one at a time. Their behaviour is simulated until they enter a region called the “black body”, which is a sphere of radius 100,000 cm around the centre of the axes, or until their energy falls below a certain threshold, below which it is reasonable to consider their behaviour negligible.

4.2 Source

As an initial case, a spherical positron source is considered, with a constant initial kinetic energy distribution equal to 300 keV, which is approximately half of the maximum energy for Fluorine-18 decay. A significant simplification is therefore introduced, namely the absence of other positron or photon sources except for the object of interest. As a result, there will be no detections of events unrelated to the object under study. This situation clearly differs from reality, since the radiopharmaceutical is distributed throughout the whole body and other photon sources exist.

4.3 Detector Geometry

Inside the FLUKA virtual environment, a ring-shaped detector was reconstructed, composed of bismuth germanate (GSO) detectors with a parallelepiped shape. The number of detectors was varied across the different simulations.

Two main limitations of the chosen configuration should be highlighted: since only a single detector layer was implemented, only two-dimensional images can be reconstructed. Furthermore, the geometry of the individual detectors is less detailed than in a real system, since there are no grooves dividing the surface into pixels. For this reason, for each received photon, the LOR is reconstructed starting from the centre of the struck face rather than from the actual coordinates of the interaction point.

4.4 Detection

The code implemented to collect data is intended to simulate the coincidence technique used by real detectors, which associates two events if they occur within a 10 ns time window. Only the detectors whose LOR passes through the region of interest are placed in coincidence. Each element is therefore paired only with other elements located within a certain arc, rather than with all the detectors in the ring, in order to avoid storing information coming from regions of space different from the one to be analysed, namely the FOV.

The output consists of a text file containing four types of data, distributed respectively over four columns: the first two identify the detectors in coincidence, while the remaining values represent the time interval elapsed between the emission of a positron and the detection of the photons generated by it. These values are used for the TOF estimation.

In the simulation, an “absolute” arrival time of the photons at the detectors is therefore not available. Consequently, it is not possible to obtain, as an error, the simultaneous detection of gamma particles originating from the annihilation of different positrons. This phenomenon, although theoretically possible in reality, is nevertheless highly unlikely. In fact, in the situation considered, the average time between successive decays is on the order of tens of nanoseconds, so

which the very low probability of detecting multiple decays close to each other in time must be added. For this reason, the error introduced by this approximation is considered negligible.

The code returns the measured time intervals with a precision much higher than the real one, on the order of hundreds of ps. For this reason, the times will be rounded to this order of magnitude, and a sensitivity of 0.1 ns will be considered, as can be observed from the code shown in Section 3.3.

Another simplification is due to the fact that the code assumes that a photon is detected at the exact instant when it crosses the surface of a detector, regardless of the type of interaction that occurs inside it. Therefore, real cases are not considered in which, for example, gamma particles enter the detectors but leave them without being detected due to scattering events.

Furthermore, in order to be detected, a photon must have an energy between 510 and 512 keV, without any error.

4.4.1 Detectors in Coincidence

In order to ensure that only the LORs passing through the space of interest are considered, it is necessary to define a FOV and make some geometrical considerations:

if the radius of the detector system is 20 cm and the radius of the FOV is 7.5 cm, only the LORs forming an angle, with respect to the diameter passing through the first detector, of $\arcsin(\frac{\pm 7.5}{20}) = \pm 22^\circ$ with respect to the line passing through the first detector and the centre must be considered.

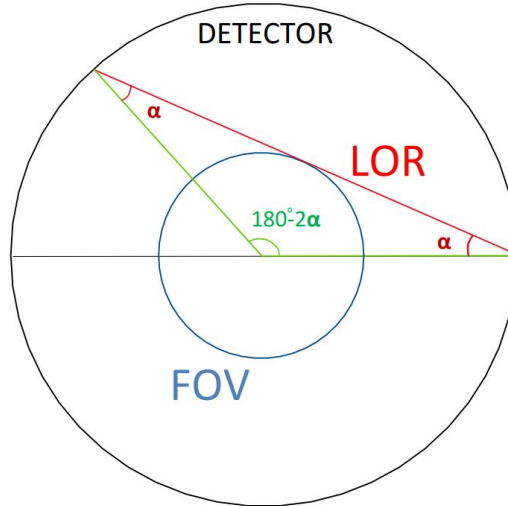


Figura 11: As can be observed, the angle identified by the two detectors and the centre must be $180^\circ - 2\alpha$, where α is the angle between the line passing through the two detectors and the diameter passing through the first detector.

Therefore, for the geometry of the simulation, the central angle of the second detector with respect to the first must be between $180^\circ - 2 \times 22^\circ = 136^\circ$ and $180^\circ + 2 \times 22^\circ = 224^\circ$.

4.4.2 Coincidence Unit Code

The code used for the detections is a user routine developed in FORTRAN.

```
ENTRY BXDRAW ( ICODE, MREG, NEWREG, XSCO, YSCO, ZSCO )  
  if (MREG == 2 .and. NEWREG>3 .and. JTRACK == 7 .and. ETRACK<0.000512 .  
    → and. ETRACK>0.000510) then  
    if (IN1==0) then  
      photon_counter = 1  
      IN1=NEWREG  
      TIME1=ATRACK  
    end if  
    if (IN1>0 .and. abs(NEWREG-IN1)>272 .and. abs(NEWREG-IN1)<448) ]  
      → then  
        photon_counter = photon_counter + 1  
        if (photon_counter==2) then  
          IN2=NEWREG  
          TIME2=ATRACK  
        end if  
      end if  
    end if  
  end if  
RETURN
```

The script shown above is used for the geometry with 720 detectors and is read after any particle crosses a surface separating two bodies.

The variables useful for the simulation, read at each crossing, are:

- MREG, merging region: the region left by the particle.
- NEWREG, new region: the region entered by the particle.
- JTRACK, particle code: defines whether the particle is a photon, neutron, etc.
- ETRACK, particle energy.
- ATRACK, time elapsed from the moment the primary particle was generated.

In order to detect the photons following positron annihilation, some conditions are imposed: the merging region must be the air surrounding the inspected body, whose code is 2; the new region must be one of the detectors, whose codes range from 4 to 723; the particle must be a photon, therefore JTRACK must be 7; finally, the energy must be 511 keV.

If the criteria mentioned above are satisfied, then a photon with valid energy has entered a detector. Therefore, the following part of the code is read.

- IN1, TIME1: these store, respectively, the number of the first detector crossed and the time at which it is crossed. IN1 is defined at the beginning of the code with an initial value equal to 0. If it is equal to 0, meaning that no value has yet been stored, it is reassigned with the value of NEWREG. If it has a value different from 0, it will no longer be modified.
- photon_counter: this variable is updated every time a detector is stored, in order to count the number of detected photons.
- IN2, TIME2: these store, respectively, the number of the detector in coincidence with the first one and the time at which it is crossed. The conditions required for the second detector to be valid are the following: IN1 must be greater than zero, since this means that the first

detector has already been stored; $|NEWREG - IN1|$ must be between 272 and 448⁵ (136 and 224 for 360 detectors; 68 and 112 for 180 detectors), so that the LOR between the detector already stored and the newly crossed detector passes through the FOV of interest.

If these two conditions are satisfied, photon_counter is updated and becomes 2. If this parameter is 2, the value of NEWREG is stored in the variable IN2 and ATRACK is stored in TIME2.

When the effects of the primary particle subsequently end, the following code is read:

```
ENTRY EEDRAW ( ICODE )

if (photon_counter==2) then
    write(80, *) IN1, IN2, TIME1, TIME2

end if

photon_counter = 0
IN1=0
IN2=0

RETURN
```

This code writes the detector regions and crossing times to a text file and then resets the variables.

5

As anticipated in the previous paragraph, if $\alpha = \pm 22^\circ$, the second detector must be within a range between $180^\circ - 22^\circ \times 2 = 136^\circ$ and $180^\circ + 22^\circ \times 2 = 224^\circ$. Since the system is composed of 720 detectors, each detector is located 0.5° from the next one. Therefore, the valid detectors are more than $136^\circ \times 2\text{detectors}/^\circ = 272\text{detectors}$ and less than $224^\circ \times 2\text{detectors}/^\circ = 448\text{detectors}$ away.

5 Simulation Accuracy

5.1 Expected Results

In the ideal case of a sphere with radius R centred at the origin, if the diameter is smaller than the height of the detectors, namely the dimension along z , the reconstructed image, as well as its projection onto the XY plane, will be a circle with a radial behaviour proportional to the lines crossing orthogonally the plane on which the projection was made. Therefore, at the centre, the intensity would be equal to $2R$. With a displacement from the centre equal to $R \cos \alpha$, the corresponding height would be $2r$, with $r = R \sin \alpha$.

Furthermore, the projection along an axis belonging to the XY plane, for example the x axis, has a behaviour proportional to the area of the intersection between the plane orthogonal to that axis and the sphere. Therefore, at the centre, the peak is equal to $R^2\pi$. With a displacement from the centre equal to $R \cos \alpha$, the corresponding section will be equal to $r^2\pi$. The following image is provided as an example.

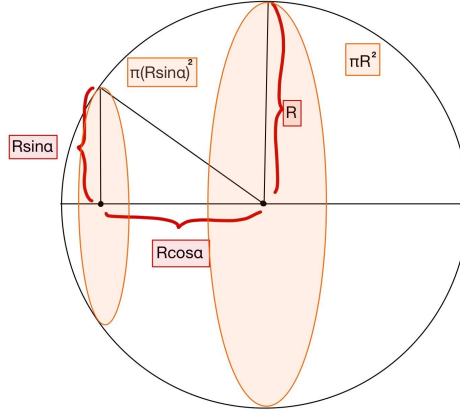


Figure 12: The image shows the behaviour of the sphere section, as well as the value of the projection along the x axis, as a function of the displacement from the centre.

Considering the expected FWHM value, the following reasoning can be applied. Half of the peak has a value equal to $\frac{R^2\pi}{2}$. This value is obtained for a displacement such that $r = \frac{R\sqrt{2}}{2}$, therefore $\sin \alpha = \frac{\sqrt{2}}{2}$ and consequently also $\cos \alpha = \frac{\sqrt{2}}{2}$. In conclusion, the displacement will be equal to $R\frac{\sqrt{2}}{2}$ in both directions, and the full width at half maximum will be $R\sqrt{2}$.

In order to obtain an image of what has been described, a MATLAB script was developed to create a sphere with radius 5 mm, project it onto the XY plane and then onto the X axis. The “ideal” results of this simulation are reported below, for different pixel sizes, so that they can be compared with the cases analysed later.

Since the size of the source is comparable to the pixel size, it can be observed that the source is not centred but shifted, and the FWHM cannot accurately reproduce the expected value, equal to $\sqrt{2} * R$, namely approximately 7 mm.

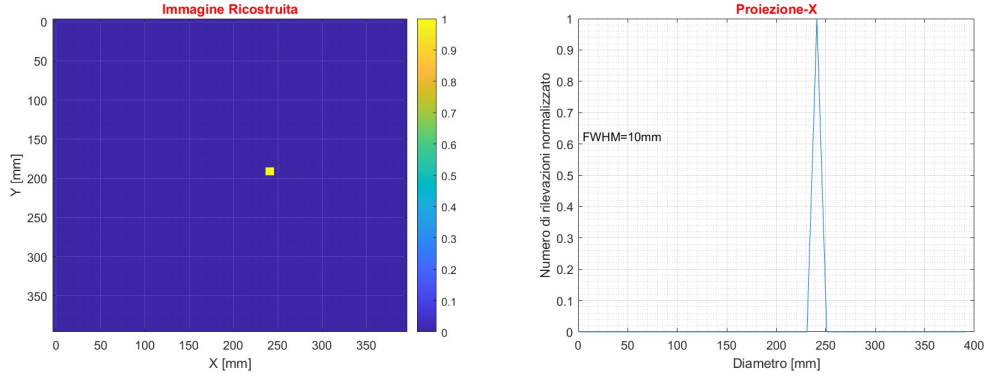


Figura 13: Image and projection along the x axis with $1 \text{ pixel}/\text{cm}^2$. $FWHM_{ideal} = 10\text{mm}$.

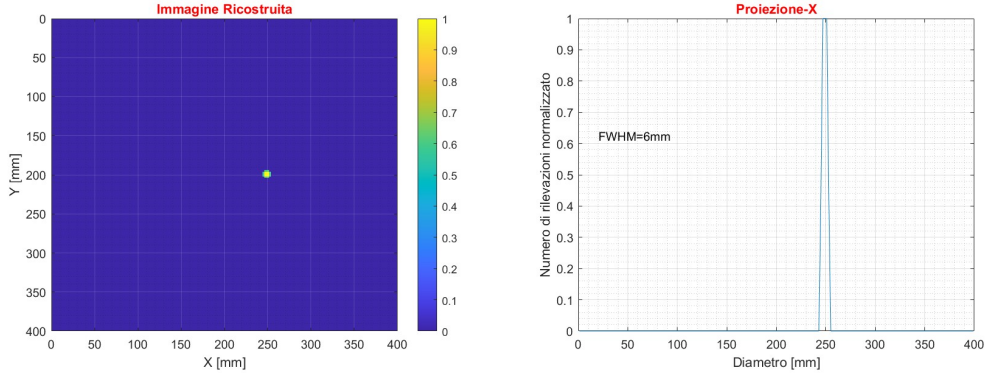


Figura 14: Image and projection along the x axis with $1 \text{ pixel}/4\text{mm}^2$. $FWHM_{ideal} = 6\text{mm}$

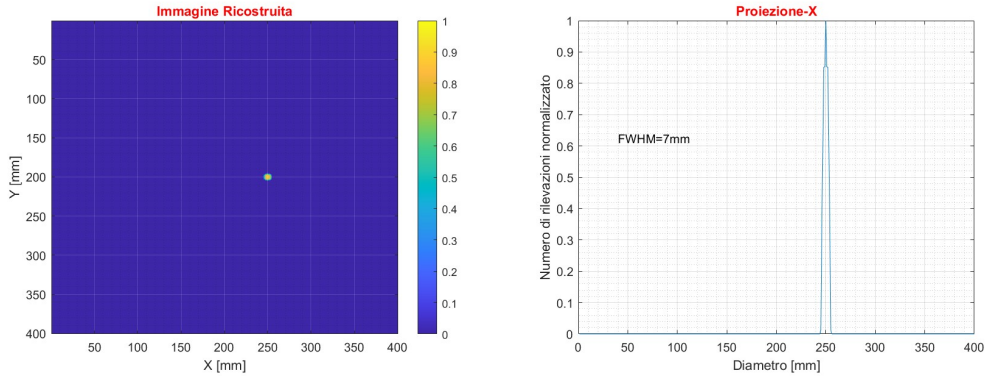


Figura 15: Image and projection along the x axis with $1 \text{ pixel}/\text{mm}^2$. $FWHM_{ideal} = 7\text{mm}$

By improving the resolution, it can be observed that the full width at half maximum of the ideal projection approaches the analytical value of 7 mm with greater accuracy.

5.2 Variables

Image reconstruction through the simulation of the PET system we created is performed using two methods: TOF and MLEM.

In order to quantify and improve the accuracy of the simulation, the variables common to both methods are:

- The number of detectors.
- The number of pixels in the image.
- The number of primaries, which is proportional to the amount of radiopharmaceutical. It should be emphasized that we will use a small number of primaries, far from the scale of a real situation. This limitation is due to the excessively high computational times required by our equipment.
- The MLEM method has an additional variable: the number of iterations of the algorithm must also be considered.

All the images shown are created from the FLUKA output, reprocessed through code written in MATLAB.

The FLUKA input is a positron source with an energy of 300 keV, approximately half of the maximum emission energy (634 keV). The source is spherical, with a radius of 0.5 cm, positioned 5 cm in the positive direction of the x axis from the centre of the axes, and inserted into a water cylinder coaxial with the z axis. The detection system consists of a ring with a radius of 20 cm, composed of parallelepiped-shaped detectors, whose number varies between 180 and 720. The detectors are 5 cm thick and deep, and wide enough to cover the entire circumference.

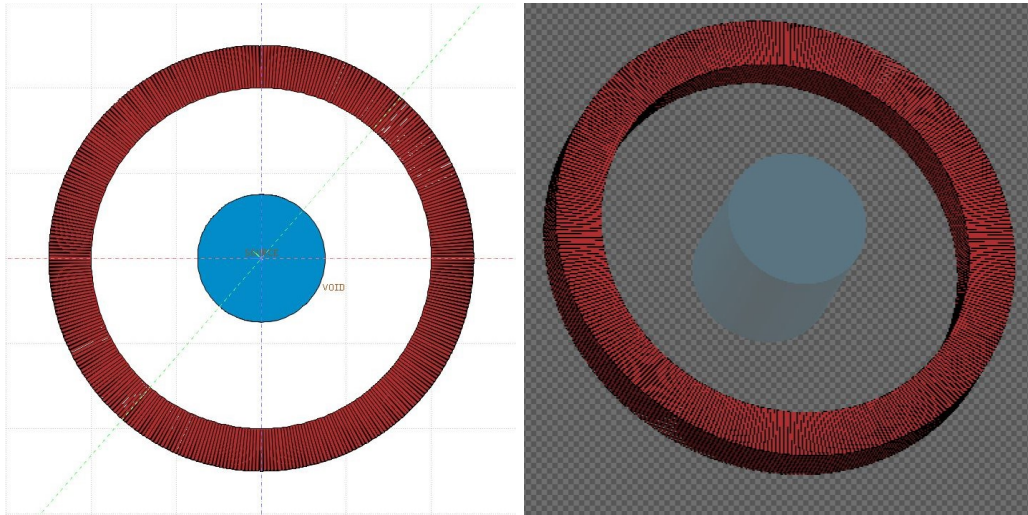


Figura 16: Geometry with 360 detectors, on the XY plane and in three dimensions.

5.3 Number of Iterations in the MLEM Method

5.3.1 Back Projection

Before analysing what happens when using the MLEM method, attention is first given to the Back Projection case. This method reconstructs the image through a simple intersection of the LORs. The thickness, or FWHM, of the projection along the x axis is much larger than 7 mm, which is the thickness that should be obtained in the ideal case starting from a source with a diameter of 10 mm. This first method does not implement any type of image filtering; for this reason, the formation of a star-shaped artefact around the sphere can be clearly observed.

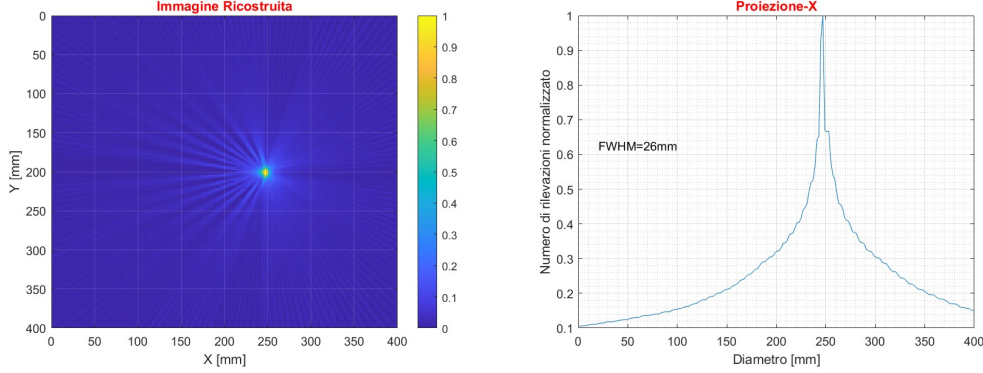


Figura 17: Image and projection along the x axis with: 180 detectors, 1,000,000 primaries, 1 $pixel/4mm^2$

5.3.2 10 Iterations

Using the MLEM algorithm with 10 iterations, it can be seen that the thickness of the peak, or FWHM, is smaller than it should be: 2 mm instead of 7 mm. However, the increase in accuracy can be immediately observed: instead of having intersecting lines, the image becomes a well-defined region, and the projection identifies the position where the source is located, although slightly shifted because the system is discrete.

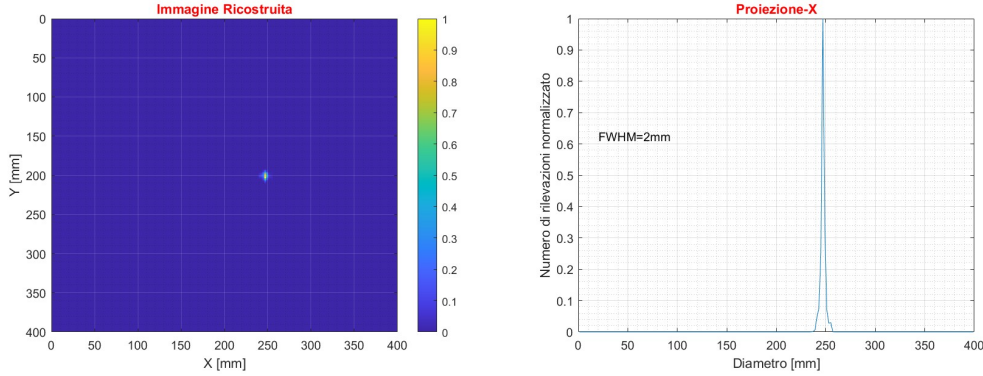


Figura 18: Image and projection along the x axis with: 180 detectors, 1,000,000 primaries, 10 MLEM iterations, 1 $pixel/4mm^2$

5.3.3 100 Iterations

Using the MLEM algorithm with 100 iterations, it can be observed that the image and the peak thickness are essentially unchanged with respect to the case with 10 iterations. However, the background noise at the base of the peak decreases, making the peak easier to identify.

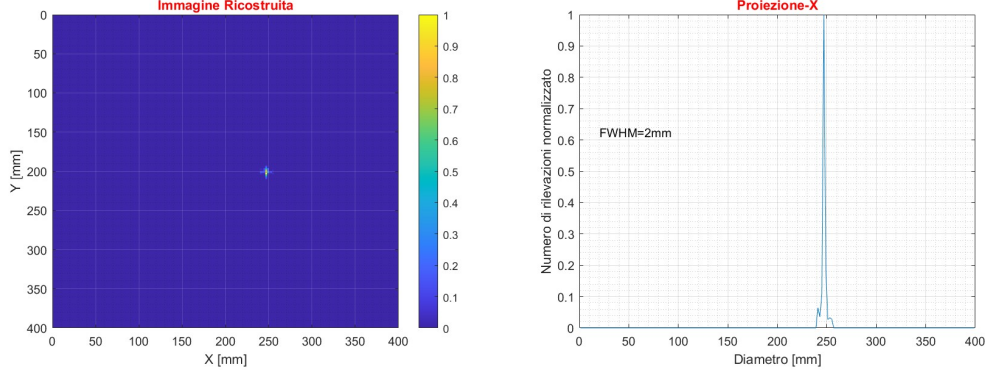


Figura 19: Image and projection along the x axis with: 180 detectors, 1,000,000 primaries, 100 MLEM iterations, $1 \text{ pixel}/4\text{mm}^2$

5.3.4 1000 Iterations

Using the MLEM algorithm with 1000 iterations, an image and a projection indistinguishable from the case with 100 iterations are observed. For this reason, the analysis will continue using 100 iterations of the algorithm for the MLEM method.

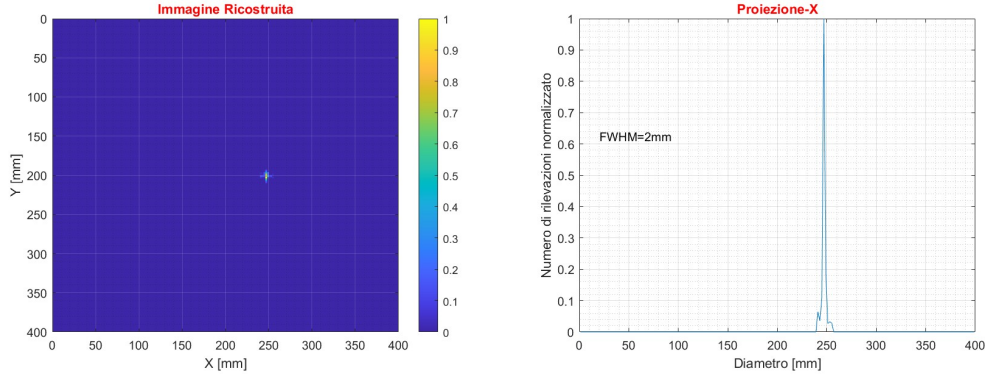


Figura 20: Image and projection along the x axis with: 180 detectors, 1,000,000 primaries, 1000 MLEM iterations, $1 \text{ pixel}/4\text{mm}^2$

5.4 Number of Primaries

It is reasonable to assume that the greater the number of primaries, and therefore the number of detections, the greater the accuracy of the reconstruction will be.

In the following paragraphs, the two reconstruction methods, MLEM and TOF, will be compared as the number of primaries varies.

5.4.1 10,000 Primaries

With 10,000 primaries, it can be observed that the peaks are still far from the ideal case: at the base of the MLEM peak there is noise, while the base of the TOF peak is too wide.

Furthermore, the thickness of both peaks differs from the expected value, namely the 6 mm of the ideal case. This value is not 7 mm because, when using $1 \text{ pixel}/4\text{mm}^2$, the ideal FWHM was also equal to 6 mm.

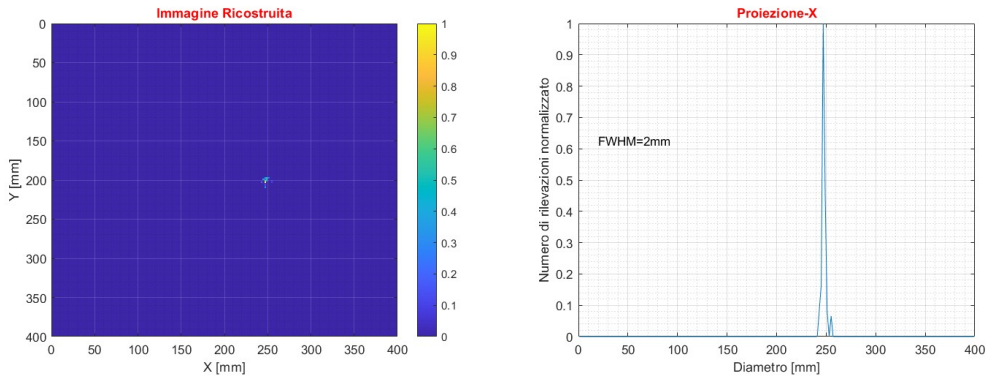


Figure 21: Image and projection along the x axis with: 180 detectors, 10,000 primaries, 100 MLEM iterations, $1 \text{ pixel}/4\text{mm}^2$

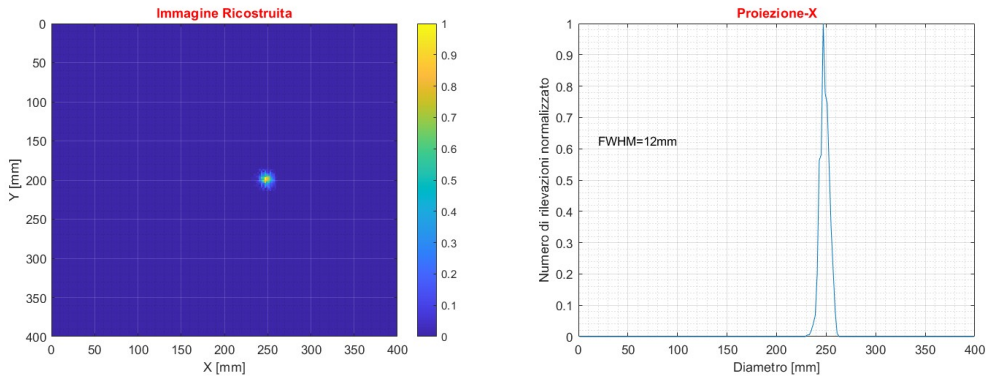


Figure 22: Image and projection along the x axis with: 180 detectors, 10,000 primaries, TOF method, $1 \text{ pixel}/4\text{mm}^2$

5.4.2 100,000 Primaries

The observations for the case with 100,000 primaries are analogous to those made for 10,000 primaries. Furthermore, no significant improvement in the images can be observed.

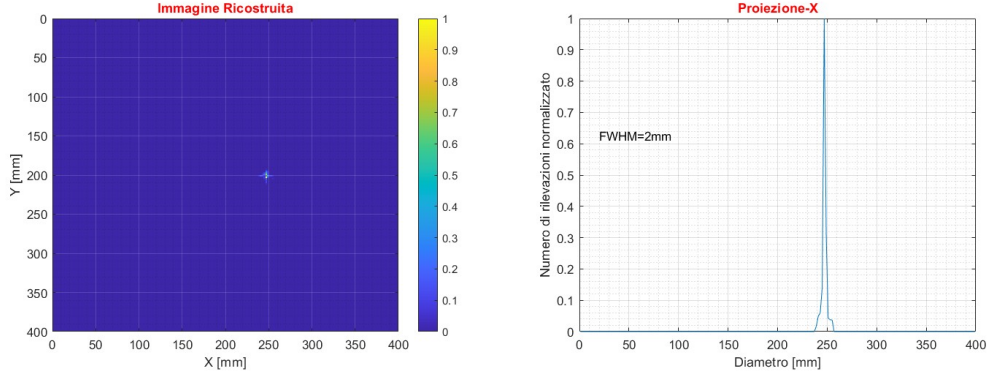


Figura 23: Image and projection along the x axis with: 180 detectors, 100,000 primaries, 100 MLEM iterations, 1 $pixel/4mm^2$

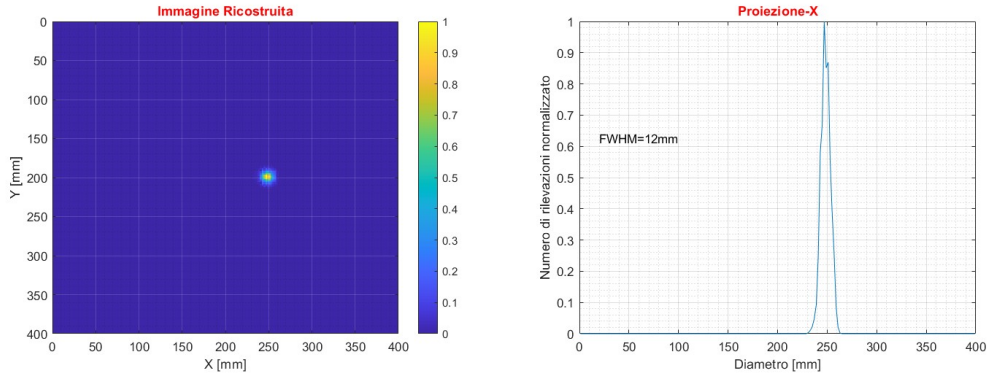


Figura 24: Image and projection along the x axis with: 180 detectors, 100,000 primaries, TOF method, 1 $pixel/4mm^2$

5.4.3 1,000,000 Primaries

By increasing the number of primaries from 100,000 to 1,000,000, no significant difference is observed for a simplified case such as the one under consideration. For this reason, it is useful to continue comparing the remaining variables without further modifying this quantity, and only at the end, when drawing conclusions, to increase the number of primaries as much as possible.

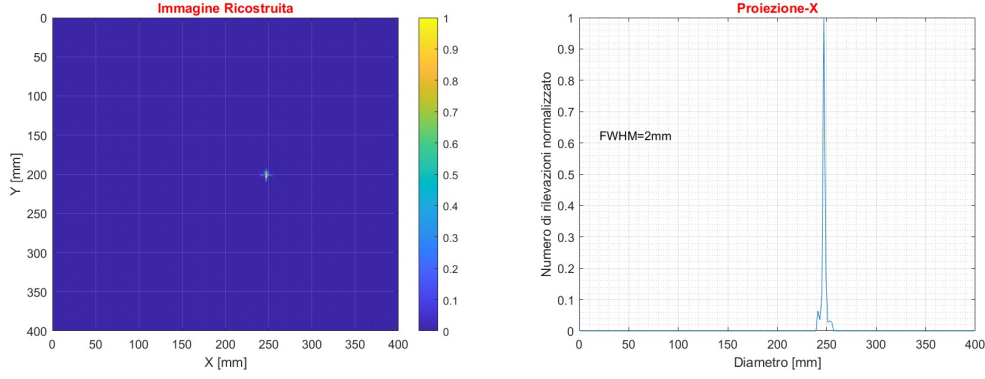


Figura 25: Image and projection along the x axis with: 180 detectors, 1,000,000 primaries, 100 MLEM iterations, $1 \text{ pixel}/4\text{mm}^2$

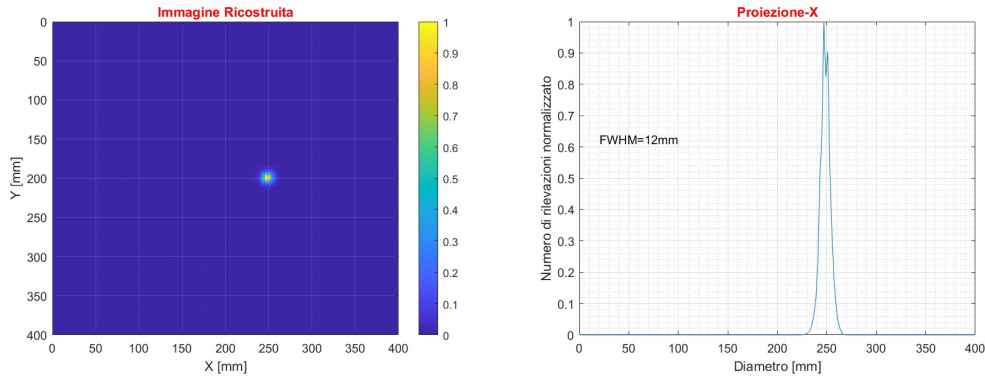


Figura 26: Image and projection along the x axis with: 180 detectors, 1,000,000 primaries, TOF method, $1 \text{ pixel}/4\text{mm}^2$

5.5 Number of Detectors

The analysis now continues by comparing the images obtained while varying the number of detectors in the ring, in order to obtain increasingly precise LORs.

5.5.1 180 Detectors

The first configuration, namely a ring of 180 detectors, was previously used in the analysis of the number of primaries. It can be observed that the full width at half maximum of the peak for both methods differs significantly from the ideal value of 6 mm.

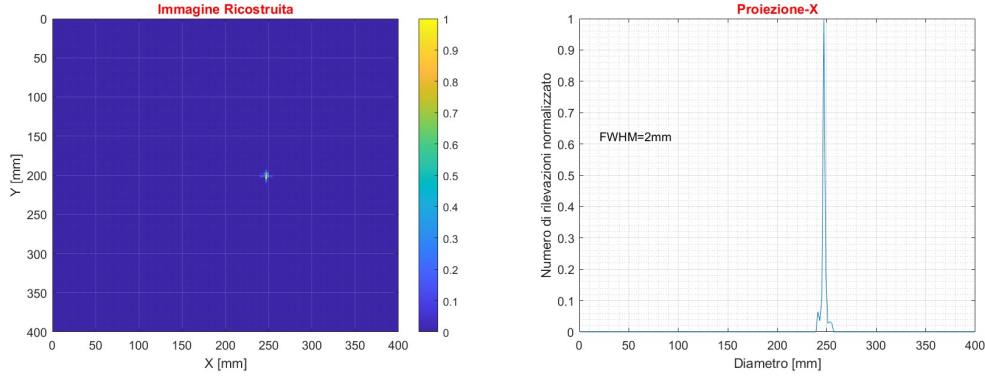


Figura 27: Image and projection along the x axis with: 180 detectors, 1,000,000 primaries, 100 MLEM iterations, 1 $pixel/4mm^2$

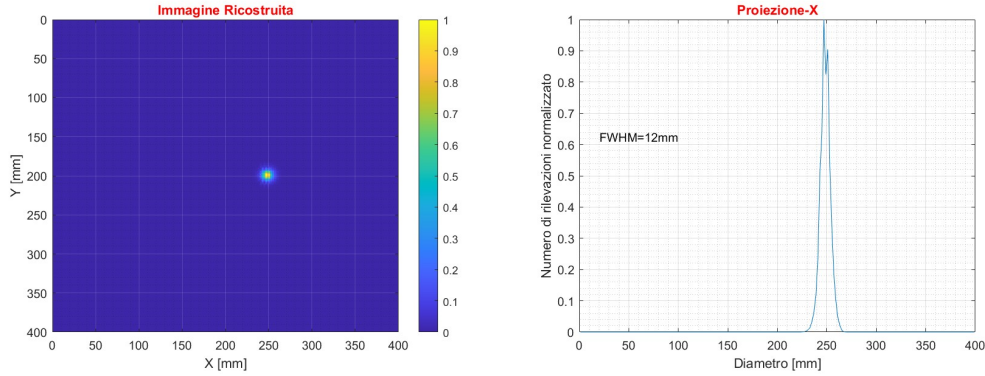


Figura 28: Image and projection along the x axis with: 180 detectors, 1,000,000 primaries, TOF method, 1 $pixel/4mm^2$

5.5.2 360 Detectors

By increasing the number of detectors in the system to 360, a significant improvement can be observed, especially for the MLEM method, for which the peak width increases considerably until it matches the 6 mm of the ideal case. On the other hand, TOF also improves, but still shows an error of 4 mm.

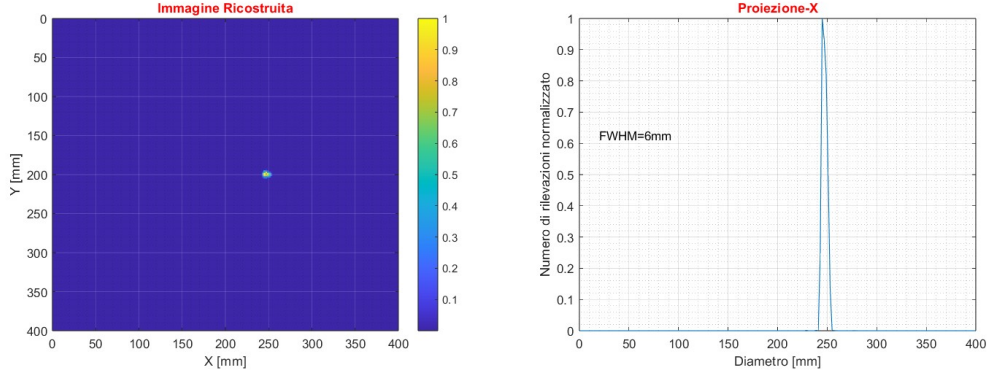


Figura 29: Image and projection along the x axis with: 360 detectors, 1,000,000 primaries, 100 MLEM iterations, 1 $pixel/4mm^2$

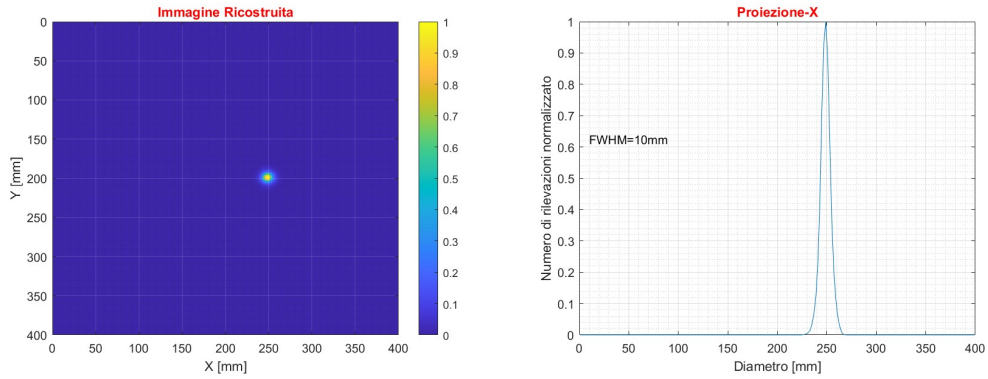


Figura 30: Image and projection along the x axis with: 360 detectors, 1,000,000 primaries, TOF method, 1 $pixel/4mm^2$

5.5.3 720 Detectors

With a further increase in the number of detectors, the improvements introduced are less significant than in the previous case, for both methods. Regarding the MLEM method, this was expected, since with 360 detectors the obtained image and projection are indistinguishable from the ideal case (Section 5.1). For this reason, when using 1 *pixel*/ $4mm^2$, the number of detectors is not the limiting variable.

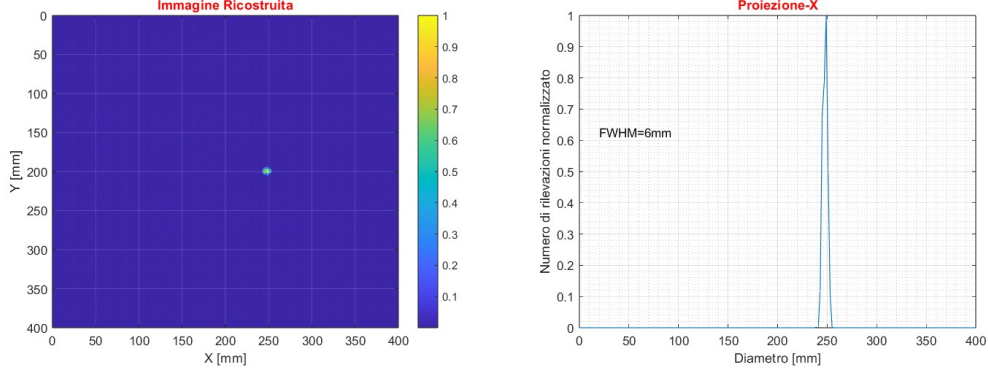


Figura 31: Image and projection along the x axis with: 720 detectors, 1,000,000 primaries, 100 MLEM iterations, 1 *pixel*/ $4mm^2$

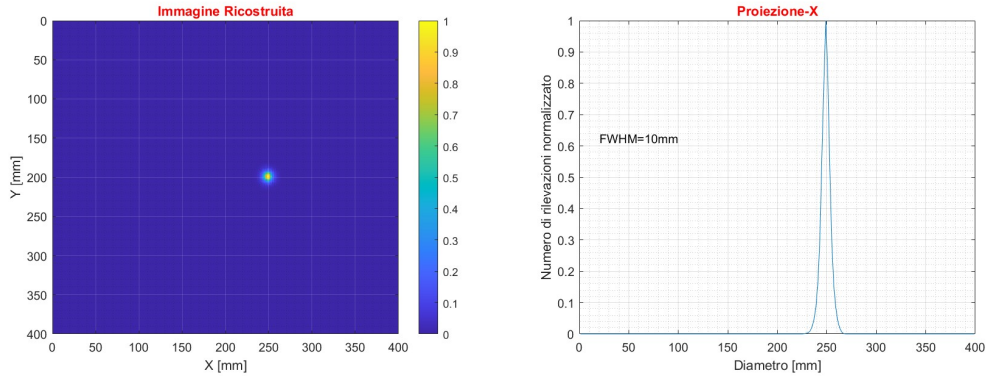


Figura 32: Image and projection along the x axis with: 720 detectors, 1,000,000 primaries, TOF method, 1 *pixel*/ $4mm^2$

5.6 Pixel Size

In a real system, the pixel size is related to the geometry of the block detectors. To simplify the study, the detectors are simple parallelepipeds and the LOR is “drawn” starting from the centres of the struck faces. The variation in image accuracy is therefore analysed as the pixel size varies between 1cm^2 and 1mm^2 .

5.6.1 $1\text{pixel}/\text{cm}^2$

For both methods, it is immediately observed that using pixels whose side length is equal to the radius of the analysed object gives inaccurate results. For example, the projection of the sphere onto the plane appears with a very different geometry, and the detection peak is shifted with respect to the real centre because of such a coarse discretization.

The results regarding the FWHM value are justified by the fact that, for the considered pixel size, this value can only vary in steps of 10 mm, and therefore cannot reproduce the theoretically expected values.

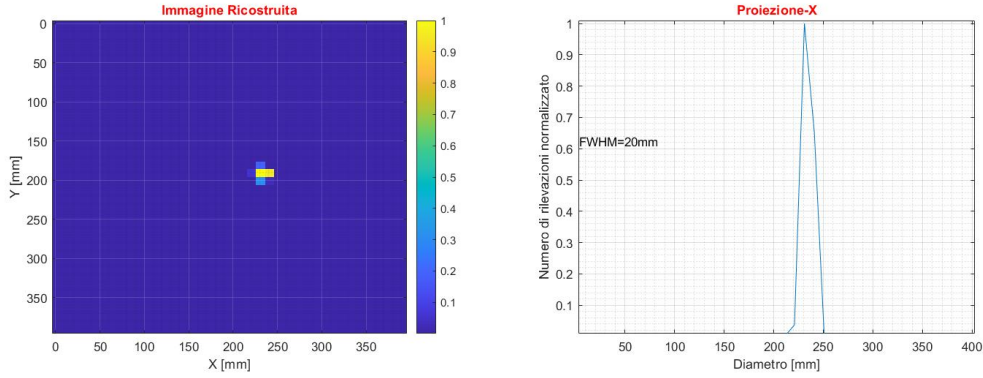


Figura 33: Image and projection along the x axis with: 360 detectors, 1,000,000 primaries, 100 MLEM iterations, $1\text{ pixel}/\text{cm}^2$

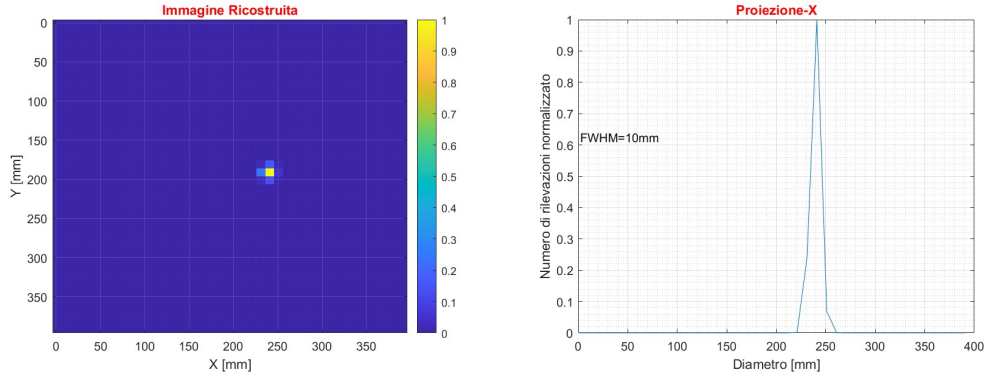


Figura 34: Image and projection along the x axis with: 360 detectors, 1,000,000 primaries, TOF method, $1\text{ pixel}/\text{cm}^2$

5.6.2 $1\text{pixel}/4\text{mm}^2$

By reducing the pixels to dimensions significantly smaller than those of the sphere, the obtained images reproduce the real geometry, and for both methods a detection peak is obtained around the centre of the sphere.

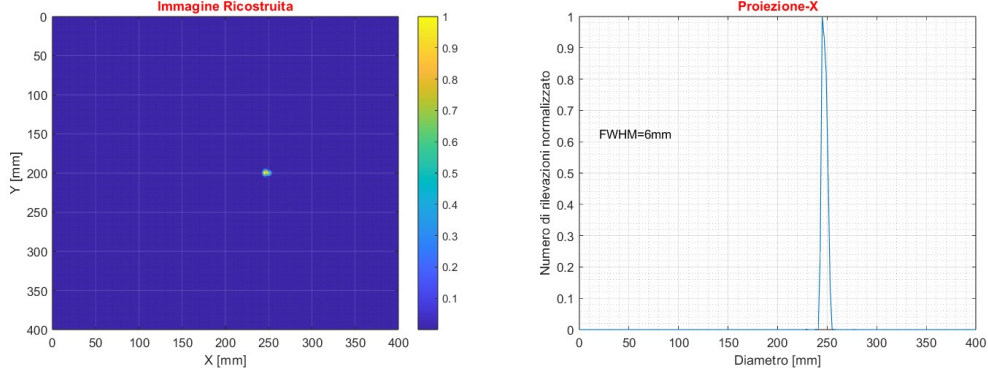


Figura 35: Image and projection along the x axis with: 360 detectors, 1,000,000 primaries, 100 MLEM iterations, $1\text{pixel}/4\text{mm}^2$

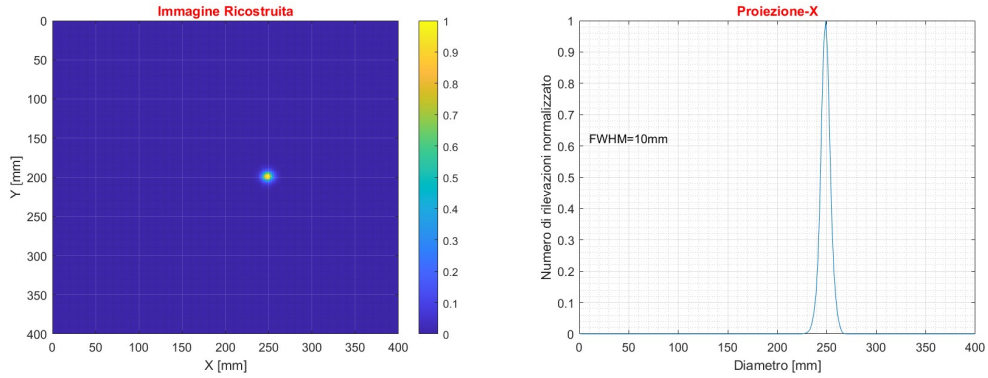


Figura 36: Image and projection along the x axis with: 360 detectors, 1,000,000 primaries, TOF method, $1\text{pixel}/4\text{mm}^2$

5.6.3 1 pixel/mm^2

With a further reduction in pixel size, the ideal full width at half maximum of the peak is 7 mm (see Section 5.1). The number of values obtained from the simulation is not sufficient to provide a uniform distribution within the geometry of the sphere; for this reason, multiple peaks appear in the detection using the MLEM method. The peak obtained with the TOF method also appears less uniform. Nevertheless, it can be observed that the TOF peak has improved, decreasing from a thickness of 10 mm to 9 mm.

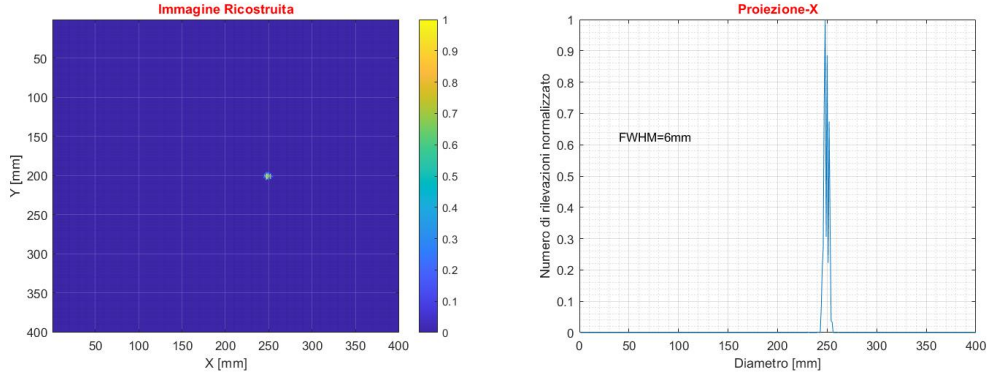


Figura 37: Image and projection along the x axis with: 360 detectors, 1,000,000 primaries, 100 MLEM iterations, 1 pixel/mm^2

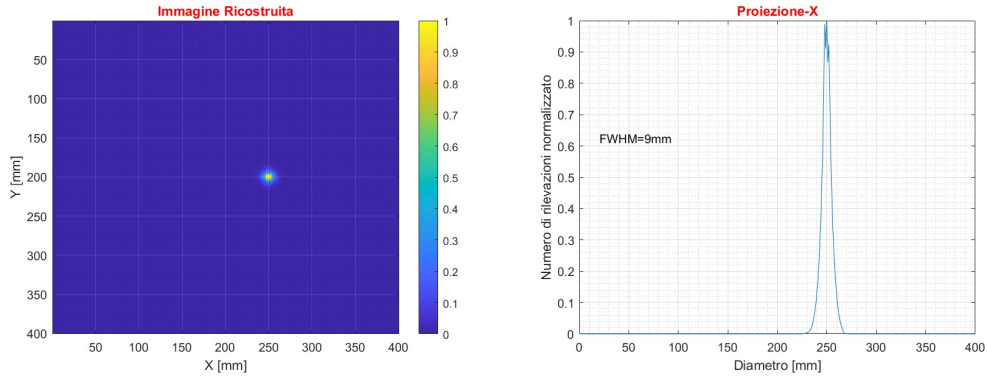


Figura 38: Image and projection along the x axis with: 360 detectors, 1,000,000 primaries, TOF method, 1 pixel/mm^2

In order to obtain a reliable projection, it would be necessary to further increase the number of primaries. However, since pixels with dimensions of 4 mm are sufficient to distinguish accurately the objects considered in the simulations, this value is considered acceptable.

An alternative that would not require excessive computational cost is to reconsider the use of 720 detectors if an image with 1 pixel/mm^2 is desired.

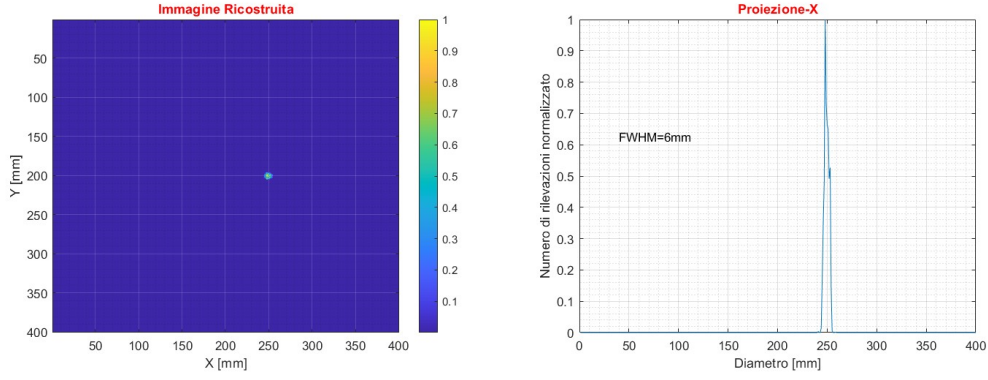


Figura 39: Image and projection along the x axis with: 720 detectors, 1,000,000 primaries, 100 MLEM iterations, 1 $pixel/mm^2$

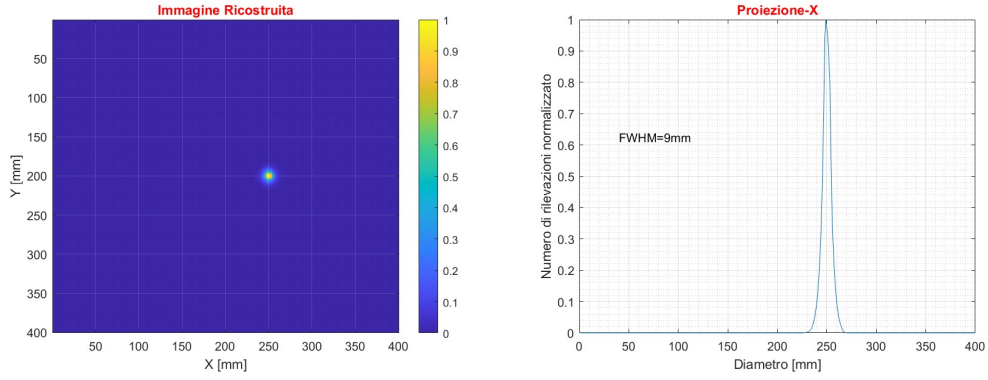


Figura 40: Image and projection along the x axis with: 720 detectors, 1,000,000 primaries, TOF method, 1 $pixel/mm^2$

It can be observed that the peaks are more uniform. Nevertheless, the changes are minimal and the FWHM values do not change. We can therefore conclude that, without further modifying the variables, the most accurate method is MLEM, with a minimum error of 1 mm on the FWHM.

5.7 “Mandala” Effect

Since the number of detectors is limited, the field of view (FOV) will be characterized by regions through which more lines of response (LORs) pass and regions through which fewer pass.

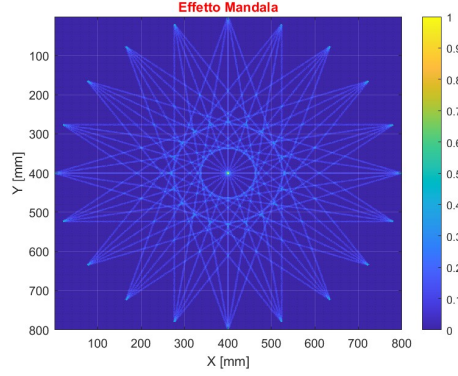


Figure 41: The LORs of a system with 20 detectors are shown, where each detector is placed in coincidence with the 5 opposite ones, in order to make the concept discussed in this paragraph easier to understand.

As shown in the figure, there are concentric rings through which more lines pass and regions through which no lines pass.

If we consider a uniform spherical distribution of sources occupying the entire FOV, each LOR will have the same count.

The reconstruction of images starting from such a distribution is then analysed; it is observed that the reconstructed image has fluctuations: the previously described mandala effect appears, although in a finer form.

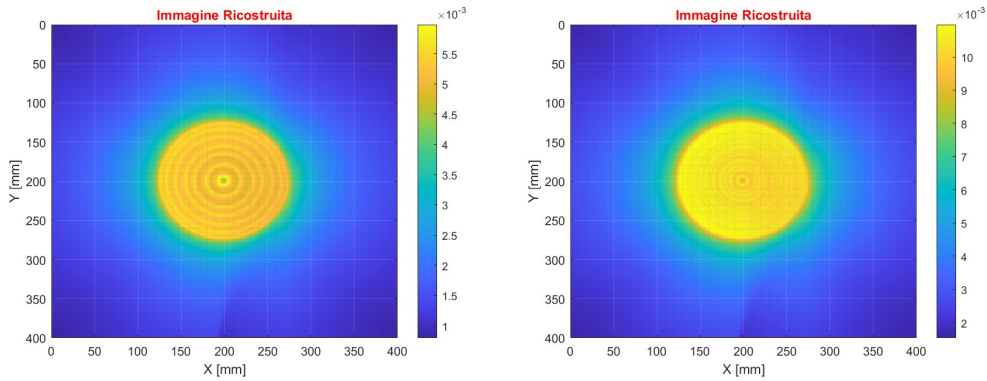


Figure 42: On the left, the combination of all the LORs with 360 detectors; on the right, with 720 detectors. Image with 1 *pixel*/4mm².

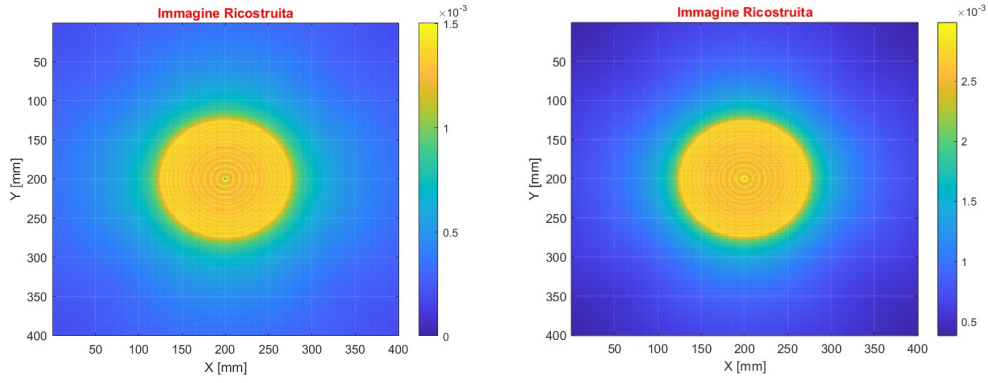


Figura 43: On the left, the combination of all the LORs with 360 detectors; on the right, with 720 detectors. Using an image with 1 *pixel*/ mm^2 .

Considering this effect, it becomes evident that using more detectors makes the grid denser, and that using 1 *pixel*/ $4mm^2$ helps reduce the regions through which no lines pass. Therefore, the best configuration appears to be the one with 720 detectors and 1 *pixel*/ $4mm^2$.

It is also necessary to investigate this effect with sources of different sizes. In fact, MLEM, being intrinsically linked to the LORs, will be more sensitive to this effect, whereas TOF, which is based on the creation of star-like lines around the centre of the source, will be less affected by it.

5.8 Effect of Source Size

It was previously shown that, for a source with a diameter of 1 cm, MLEM is more accurate compared with TOF. However, in the next two paragraphs it will be shown that, for sources with a diameter of 2 cm, the error is the same ($FWHM_{TOF} = FWHM_{ideal} + 2mm$; $FWHM_{MLEM} = FWHM_{ideal} - 2mm$), but that the shape of the peak is better in the TOF case. By further increasing the diameter, the TOF method proves to be better both in terms of shape and FWHM. For a source with a diameter of 4 cm, $FWHM_{TOF} = FWHM_{ideal}$.

5.8.1 Effect of Source Size on MLEM

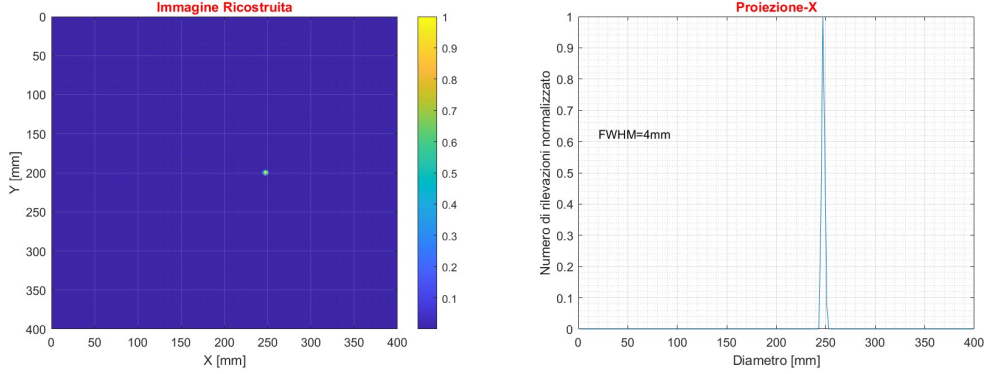


Figura 44: Image and projection along the x axis with: 360 detectors, 1,000,000 primaries, 100 MLEM iterations, 1 $pixel/4mm^2$.

Source radius 2.5 mm, therefore $FWHM_{ideal} = 3.5mm$

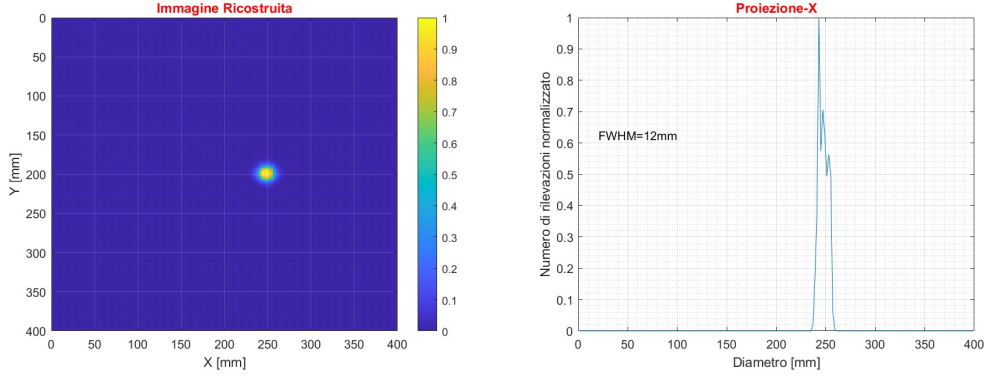


Figura 45: Image and projection along the x axis with: 360 detectors, 1,000,000 primaries, 100 MLEM iterations, 1 $pixel/4mm^2$.

Source radius 10 mm, therefore $FWHM_{ideal} = 14mm$

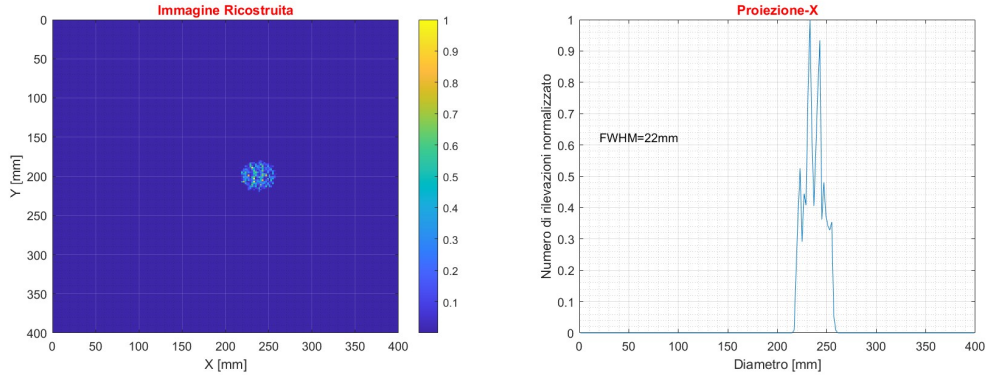


Figura 46: Image and projection along the x axis with: 360 detectors, 1,000,000 primaries, 100 MLEM iterations, 1 $pixel/4mm^2$, position $x = 4cm$. Source radius 20 mm, therefore $FWHM_{ideal} = 28mm$

It can easily be observed that, as the source size increases, this method becomes less and less accurate: the shape of the peak is less regular, the FWHM value progressively deviates more from the ideal one, and multiple peaks are formed. The increase in the number of peaks suggests that the degradation in accuracy is caused by the mandala effect mentioned previously.

5.8.2 Effect of Source Size on TOF

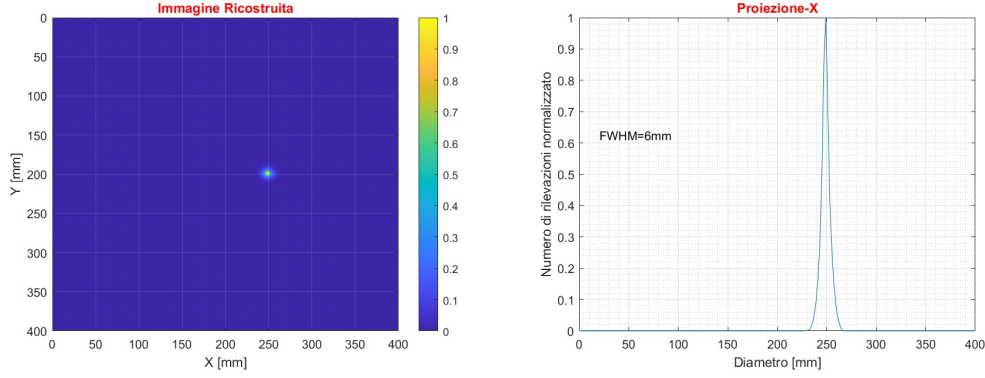


Figura 47: Image and projection along the x axis with: 360 detectors, 1,000,000 primaries, 100 MLEM iterations, 1 $pixel/4mm^2$.
Source radius 2.5 mm, therefore $FWHM_{ideal} = 3.5mm$

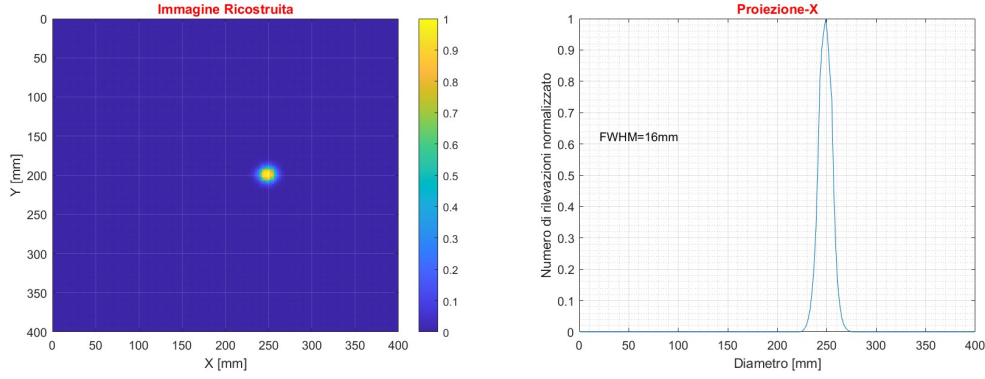


Figura 48: Image and projection along the x axis with: 360 detectors, 1,000,000 primaries, 100 MLEM iterations, 1 $pixel/4mm^2$.
Source radius 10 mm, therefore $FWHM_{ideal} = 14mm$

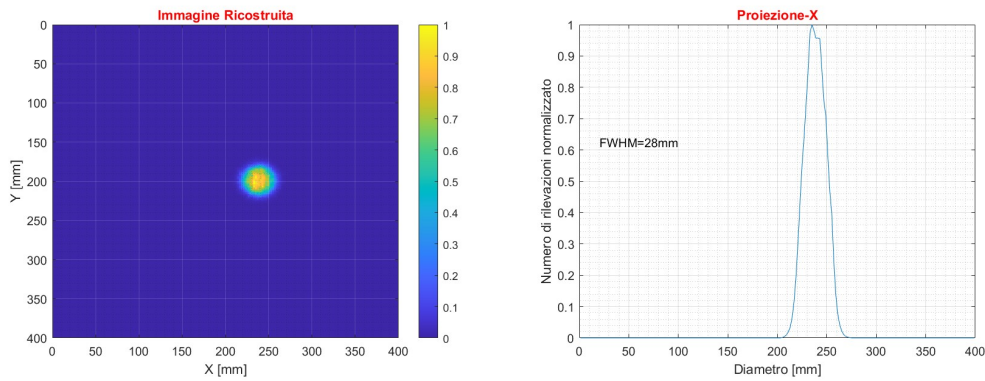


Figura 49: Image and projection along the x axis with: 360 detectors, 1,000,000 primaries, 100 MLEM iterations, 1 $pixel/4mm^2$, position $x = 4cm$.
Source radius 20 mm, therefore $FWHM_{ideal} = 28mm$

It can be observed that, with the TOF method, the peak remains almost always uniform and, as the source size increases, the FWHM gets closer and closer to the ideal one until it coincides with it. The inaccuracy for smaller sources can be justified by the fact that the error assigned to the instrument precision causes the full width at half maximum of the Gaussian used to assign probabilities to the points on the LOR around the measured one to be 15 mm. Therefore, for smaller sources, it noticeably broadens the peak width.

6 Reproduction of a Realistic Case

In this section, a realistic case will be analysed and simulated, comparing the limitations and simplifications required for our investigation with what happens in reality.

6.1 Radiopharmaceutical Administration

When 18F-FDG is administered to the patient, it distributes throughout different parts of the body.

The uptake⁶ of the radiopharmaceutical is evaluated using semi-quantitative indices, the most commonly used of which is the SUV (Standardized Uptake Value). This is a relative value that expresses the intensity of uptake. It is related to the activity administered to the patient and to the patient's body weight, and is calculated using the following formula:

$$SUV = \frac{\text{Detected activity}[\frac{\text{Bq}}{\text{mL}}]}{\frac{\text{Administered dose}[\text{mCi}]}{\text{Body weight}[\text{g}]}} \quad (3)$$

Taking into account the standard amount of activity administered to the patient during the diagnostic analysis, equal to 3.7 MBq/kg, the patient's weight, here assumed to be 70 kg, and the average SUV associated with a glioblastoma, 13.5 g/ml, the number of primaries decayed within the tumour region, which would actually be detected through PET, will be evaluated. A scan time of 15 minutes will be assumed.

To do this, it is necessary to consider the decay constant of the radionuclide, equal to $1.05 \cdot 10^{-4} \text{ s}^{-1}$, and the tumour size, here considered as a sphere with a diameter of 4 cm.

In this way, it is possible to estimate the activity measured by PET in the region of interest, namely within the tumour volume. Starting from this value, using the decay constant of 18F, the number of primaries present inside the glioblastoma during the scan can be obtained.

Through the previous analysis, the number of decays resulting in a semi-realistic case can be calculated as follows:

$$A = \frac{SUV \cdot D_s}{P} \quad (4)$$

where D_s indicates the dose administered to the patient, P the body weight, and A the total activity, from which the total number of primaries N_0 is extrapolated:

$$N_0 = \frac{A}{\lambda} \quad (5)$$

where λ represents the decay constant of 18F. According to the decay law:

$$N(t) = N_0 \cdot e^{-\lambda t} \quad (6)$$

the number of decayed primaries N_{dec} can be calculated as the difference between the total number of primaries N_0 and the number of primaries present at time t (equal to 900 seconds, the scan time):

$$N_{dec} = N_0 \cdot (1 - e^{-\lambda t}) = 1.75 \cdot 10^{12} \quad (7)$$

The number of decays is much higher than the one considered in our PET model, which is subject to simplifications that will be highlighted later.

It is also possible to make some evaluations regarding the effective dose administered to the patient, thanks to the conversion factors available in ICRP Publication 106. In the analysed case, using the conversion factor ($0.019 \frac{\text{mSv}}{\text{MBq}}$), an effective dose equal to 4.92 mSv is obtained. Taking into account the limits and principles applied in radiological protection practice, it can be stated that this effective dose is justified for the case considered.

⁶Uptake is the amount of radiopharmaceutical, and therefore of the Fluorine-18 radionuclide, that concentrates in a specific organ or tissue.

6.2 Simplifying Assumptions

The difference between the real data and the data used in the simulation is primarily due to computational limitations, which make it impossible to simulate such a high number of primaries. Furthermore, the simulation is based on a series of simplifying assumptions, which are described below.

- In the case of the brain, tumour detection can be difficult, even for large tumours, because of the high metabolic activity of the brain. This implies the presence of glucose within the brain tissue, in our case 18F-FDG. At this point, distinguishing the 18F-FDG present in the tumour becomes more complicated.

Some studies have shown, however, that this problem can be addressed by performing the PET scan several hours after the administration of the radiopharmaceutical. This is due to the different glucose excretion rates of healthy brain cells compared with damaged cells, which therefore allows the tumour mass to be better distinguished from the healthy region.

This assumption makes it possible to simplify the analysis of the case considered: it will be assumed that the time elapsed between radiopharmaceutical administration and the PET scan is sufficient to obtain a ratio between the concentrations⁷ of 18F-FDG equal to 10:

$$\frac{C_t}{C_c} = 10; \tag{8}$$

It is worth recalling the simplifications required for the use of FLUKA, as mentioned above:

- The positron energy is assumed to be constant and equal to 300 keV;
- The absence of other photon sources;
- The time associated with each photon represents the time interval between positron annihilation and detection in the scintillator;
- Detection occurs exactly at the moment when the detector is crossed, and does not depend either on the released energy or on the absence or presence of a photoelectric effect;
- Only photons with energy between 510 and 512 keV are detected;

⁷ C_t is the concentration of 18F-FDG in the tumour, while C_c is the concentration of 18F-FDG in the brain.

6.3 Simulation, Detection and Image Reconstruction

In order to obtain a simulation with a more realistic shape of the tumour mass and to take into account the distribution of the radiopharmaceutical within the brain tissue, the FLUKA “source” user routine can be used. The following simulation was performed by modelling a metastatic tumour through three spheres with diameters of 1 cm, 1.5 cm and 4 cm, respectively, positioned inside a fourth sphere with a diameter of 13 cm representing the brain.

The sources described above are in turn inserted into a water cylinder with a height of 15 cm and a diameter of 15 cm, coaxial with the detector and positioned at its centre.

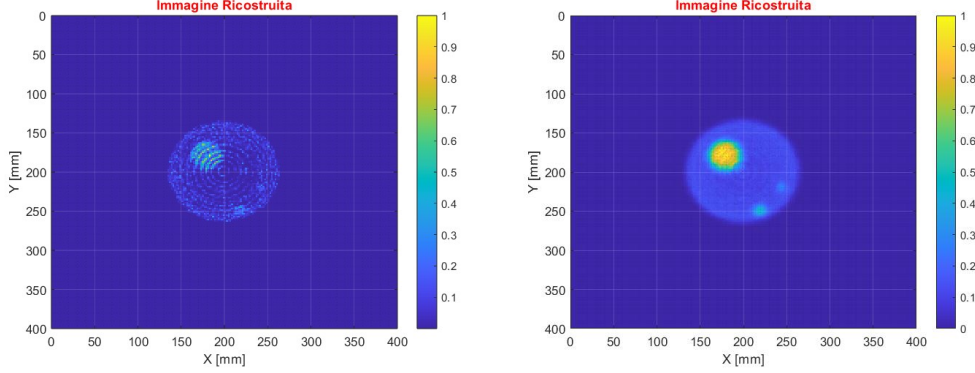


Figura 50: Images reconstructed with 720 detectors, 1 $\text{pixel}/4\text{mm}^2$, and 10,000,000 primaries. On the left, the image reconstructed with 100 iterations of the MLEM method; on the right, the image reconstructed with TOF.

Since the sources are multiple and have a diameter greater than 1 cm, which is the value for which MLEM was better than TOF and beyond which TOF has equal or higher accuracy, the MLEM method produces a more irregular image compared with TOF. The latter returns an image that, although blurred, is uniform and allows the sources to be easily identified. This qualitative consideration goes against what emerged from the quantitative evaluation of the instrument accuracy. However, it is justified by the fact that, in the presence of multiple sources, TOF makes it possible to identify a region where a source is certainly present, whereas without TOF each LOR could pass through two or more sources, providing only partial information.

6.3.1 Effect of Multiple Sources

To support the statements just made, a simulation was carried out with 1,000,000 primaries and a source composed of two spheres with radius 1 cm, positioned at ± 5 cm along the x axis.

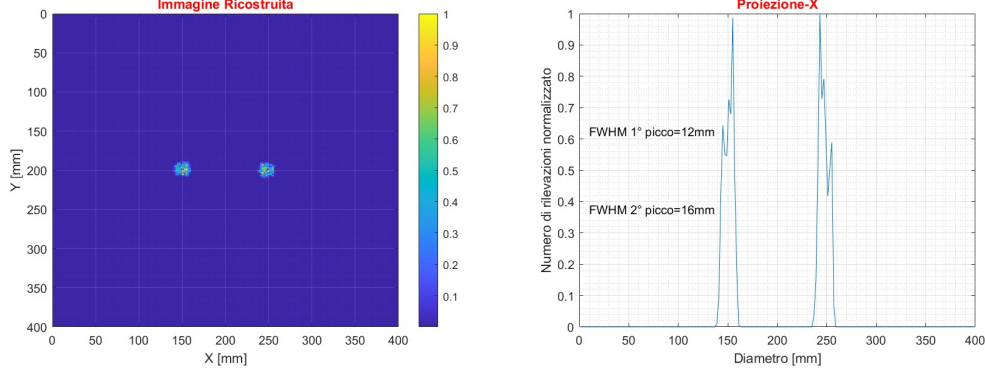


Figura 51: Images reconstructed with the MLEM method, 100 iterations, 360 detectors, 1 $pixel/mm^2$, 1,000,000 primaries.

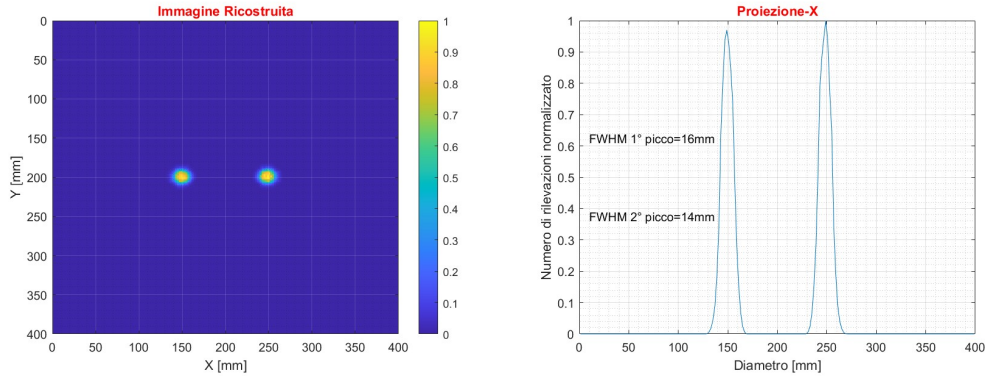


Figura 52: Images reconstructed with the TOF method, 360 detectors, 1 $pixel/mm^2$, 1,000,000 primaries.

It was previously observed (see Section 5.8) that, for spheres with radius 1 cm, the accuracy of the two methods was the same. However, it can be observed that the FWHM values obtained with the TOF method are closer to those of the ideal case, namely 14 mm (in the ideal case $FWHM = R\sqrt{2}$). This demonstrates that multiple sources degrade the image reconstructed by MLEM more than the image generated with TOF.

It is also observed that the mandala effect creates separate peaks in the reconstruction obtained through MLEM, making the image irregular.

6.4 Realistic Case

Keeping the same simplifications as before, a simulation was carried out with four spherical sources:

- 1) $R=6.5$ cm, centre coordinates: $(0 ; 0 ; 0)$
- 2) $R=0.5$ cm, centre coordinates: $(4.0 ; 2.0 ; 0)$
- 3) $R=0.75$ cm, centre coordinates: $(2.0 ; 5.0 ; 0)$
- 4) $R=2$ cm, centre coordinates: $(-2.0 ; -2.0 ; 0)$

However, instead of being placed inside a water cylinder, these spheres are located in a geometry that accurately reproduces a head.



Figure 53: Section on the XY plane.

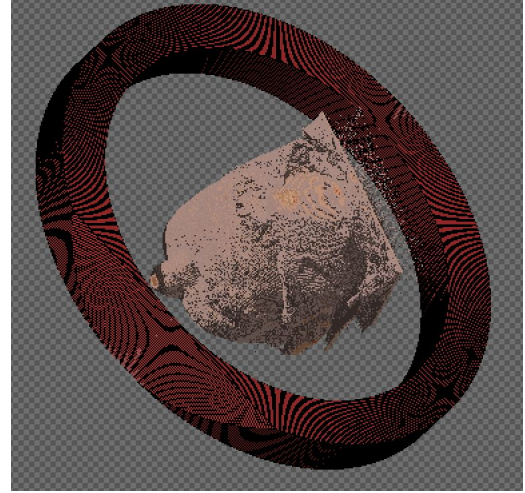


Figure 54: Head positioned at the centre of the detector.

Compared with the case involving multiple sources in a cylinder, the image is expected to be more disturbed.

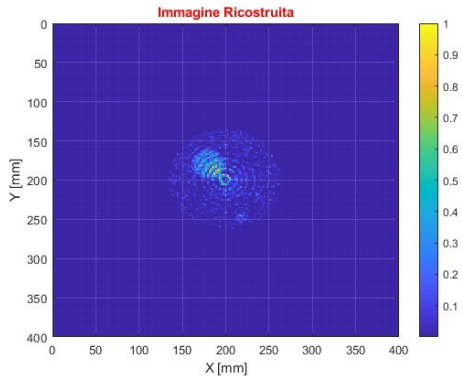


Image reconstructed with MLEM, 100 iterations.

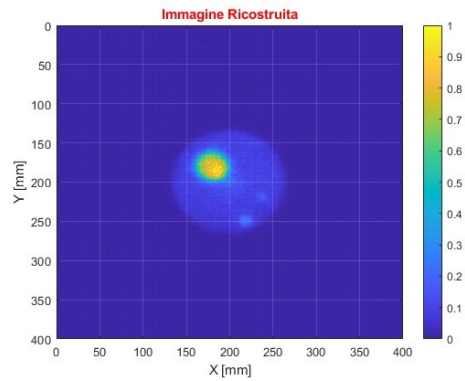


Image reconstructed with TOF.

Figure 55: 720 detectors, 1 *pixel*/4mm², 10,000,000 primaries.

As expected, the images are less clear. In fact, only the two largest sources are clearly distinguishable, while the two smaller ones are difficult to see. With MLEM, the smallest source is impossible to distinguish. With TOF, although very blurred, the smallest source can still be identified.

7 Final Remarks

The simulated model includes several simplifications, including: constant positron energy, no photon sources except those generated by positrons, absence of an absolute time, detection at the simple crossing of the detector surface, and detection only of photons with energy between 510 and 512 keV.

Despite these simplifications, it was possible to obtain a reliable reconstruction of the source. Both the MLEM and TOF methods have spatial resolutions with deviations, between the obtained and ideal full width at half maximum, ranging between 1 and 2 mm.

MLEM generally proves advantageous in terms of absolute accuracy, but TOF reconstructs multiple sources and larger images with greater accuracy. Therefore, no remarkable performance gap is observed. However, if it were possible to increase the temporal sensitivity of the detectors, TOF would certainly be the better solution. Furthermore, it should be considered that the image obtained using the TOF method was not filtered in any way.

To improve the simulation, some changes could be applied:

- By generating a random number and knowing the energy distribution of positrons emitted by Fluorine-18, a different energy should be assigned to each positron;
- Other single photon sources should be added. In fact, other emitters exist, and after β^+ decay, Fluorine can be left in an excited state and emit a γ ray;
- Knowing the decay law, an absolute time could be added to each individual time. This operation would improve the simulation only if the number of positrons were drastically larger. With 1 or 10 million primaries, the time window between one annihilation and the next is on the order of tens of nanoseconds. Therefore, considering that more than 90% of positrons emit gamma rays that are not detected, having absolute times so far apart would be exactly equivalent to having relative times;
- Detection should occur only if the photon deposits energy in the block detector. In the case of scattering, namely when energy is deposited in multiple detectors, their positions should be averaged using the deposited energy as a weight.

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